

BEHAVIOURAL SAFETY

8.6.2.5 Review critical behaviours

A review of critical behaviours can be carried out within a defined timescale to ensure that the programme is still addressing areas that are relevant and improving. Once these behaviours have become habitual they may be replaced by other significant critical behaviours, which are developed through the same looped cycle as detailed by this general model.



Once implemented, the success of the programme is highly dependent upon the commitment made, particularly by managers, at each stage of the process. People will value action and communication regarding progress and assume a lack of commitment if these are not carried through as expected.

8.7 Reducing human error and influencing behaviour

People can cause or contribute to accidents by failing to do a job correctly, and by hearing but not listening to or understanding health and safety information associated with the task, thereby failing to work to expectations. People do not generally set out to make errors deliberately, but they are often set up to fail by the way the brain processes information. For example, errors may occur as a result of the following.

- Stress or fatigue.
- Working long hours without sufficient rest/erratic shift patterns.
- Slipping into autopilot mode.
- A lack of training.
- Poor design of equipment.
- Poorly maintained equipment.
- Unclear procedures, or lack of effective procedures.
- Shortcomings in the culture of the organisation.
- Poor decision making, even when aware of the risks.
- Misinterpretation of a situation or inappropriate action taken.
- Poor situational assessment leading to incident escalation.
- Bored or disheartened staff.
- Medical problems.

When confident and knowledgeable to do so, other people (such as colleagues, supervisors and managers) can intervene to prevent potential accidents or mitigate their possible effects. The severity of an accident can be reduced by the effectiveness of the immediate or emergency response. This effectiveness can be improved by planning and appropriate training. The following are some of the topics that can influence human error and behaviour.

8.7.1 Active and latent failures

The consequences of human failure can be immediate or delayed. It is important to have an understanding of active and latent failures and how they affect health and safety.

Active failures have an immediate consequence and are usually made by frontline people (such as drivers, operators or even the public). In a situation where there is no room for error, these active failures have an immediate impact on health and safety.

Latent failures are caused by actions generally involving people (such as designers, decision makers and managers) whose tasks are removed in time and space from operational works. Latent failures are typically failures in the design, implementation or monitoring of health and safety management systems.

Latent failures can provide a significant potential danger to health and safety. These can be highlighted only through positive safety discussions that utilise the experience and knowledge of the workforce. Latent failures (such as the examples on the next page) are generally hidden within an organisation until they are triggered by an event likely to have serious consequences.

- Poor design of workplaces, plant and equipment.
- Gaps in supervision.
- Undetected manufacturing defects.
- Maintenance failures.
- Unworkable procedures.
- Clumsy automation.
- Ineffective competency assurance.
- Ineffective training.
- Ineffective communications.
- Uncertainties of role and responsibility.
- Ageing assets, plant, tools and equipment.
- Poor planning (insufficient people/time).
- Poor information on health and safety incidents.



*Latent error**

** The image of the latent error (right) shows graphite blocks fitted in the Hinckley Point B reactors. These blocks have small cracks within them, which has resulted in the reactors being run at 70% power output. It is believed that these cracks could be due to handling the blocks during construction. This has not affected safety but it has affected productivity. This is a perfect example of a latent error.*

8.7.2 Impact of change

Change of any type within an organisation can create people issues and behavioural issues. For example:

- an organisation may appoint new leaders and managers
- the need to develop new skills and capabilities may increase.
- roles can change

Workers may be uncertain and resistant because of the following.

- Limited communication and explanation of changes.
- They feel that they will be disadvantaged by the change.
- People do not see the need for change.

Dealing with these issues on a reactive, case-by-case basis puts the progress of the job, workforce morale, and overall performance of the behavioural approach at risk.

Change is unsettling for people at all levels of an organisation. The manager, or management team, will need to focus on working together and understanding that change can lead to alterations in typical working practice, stressful responses and a potential need for increased communication and support. Change affects people personally; this can also be a factor in the implementation of a behavioural safety approach. Individuals (or teams of individuals) will require timely information regarding the following.

- The changes they will see as a result of a behavioural safety programme.
- How they and their behaviours will be measured.
- What will be expected of them during and after the change programme.
- What success or failure will mean for them and those around them.

8.7.3 Leading by example

The implementation of behavioural safety can pose particular problems with a fragmented and mobile workforce, such as that found in the construction industry. To be successfully implemented on site, the principles of behavioural safety will need to be embedded within the organisation's culture and understood by the workforce and management from the initiation and planning stages.

The foundations, expectations and compliance processes should be communicated, discussed and agreed from the beginning. If employees and contractors are involved, empowered to influence and are provided with the relevant induction, critical safety behavioural standards will be set for the future.

Managers are a critical element within the success or failure of programmes. Non-compliance and the breaking of basic rules (such as not wearing safety helmets or safety goggles) sends a message that a behavioural safety approach is not taken seriously, despite policies and procedures to the contrary. Similarly, when senior managers visit sites, they should receive the same induction and live by the same rules – body language and leading by example can send a powerful message.

8.7.4 Communication style and content

Communication style and content are important elements that contribute significantly to the building of trust. People feel trust when an individual's behaviour and body language is congruent with what is being said – when 'walking the talk' is evident. Our brains are particularly good at picking up on communication signals from others, whether intended or, as can sometimes be the case, unintended.

Communication is at the heart of all that we do, at work and in our own time. It takes many forms and can be transmitted via various media (for example, face-to-face, radio, telephone, email or video conferencing). It is essential, especially within our working environment, that we get it right. Barriers to effective communication may include background noise, the type of language used and sociocultural issues. The potential for confusion and misinterpretation can be high. In any communication process, it is vital to give the person receiving information the time and space to be able to think and formulate a response. In communication it is the quality, not the quantity, that matters.

8.7.4.1 Verbal and non-verbal communication

During a normal conversation, we usually transmit and receive in three ways.

- What is said (words).
- Body language (conscious or not).
- How it is said (tonality).

During periods of high workload or stress, our body language goes largely unnoticed. This is when the words we use and the way in which we say them become more important. In addition, our listening capability reduces as our workload or stress increases.

- Communication involves both a communicator and a receiver and is generally the responsibility of the communicator.
- What we say, how we say it and when we say it are perceived differently by different individuals.
- Overload of work and stress can be main factors that hinder effective communication. If the receiver is feeling overloaded, communication will need to reflect this. If the message is important then we need to lessen the workload.
- Job-critical communication requires undivided attention by both the communicator and the receiver.
- The most effective communicators will always check that the meaning they intend is the one that has been received.

There is a difference between hearing and listening. Hearing is a mechanical process involving the way sound waves are translated.

BEHAVIOURAL SAFETY

Listening is paying attention to what is being said. When we do this, we are able to evaluate the message within the correct context. An appropriate response can then be planned.

Listening is often described as taking place at different levels, from the less attentive Level 1 (conversational listening) to the more engaged Level 4 (deep listening). At this level, the listener is paying as much attention to body language and signals regarding what is not being said as they are to the words spoken. This type of listening takes skill and practice, and is the mark of a good communicator. Observers within a behavioural safety programme would find great benefit in becoming listening masters, so they have the potential to pick up on, and challenge, safety-related information that could otherwise be missed.

8.7.4.2 Questioning skills

How questions are asked can inform the overall effectiveness of a discussion in terms of information shared. There are several types of question, with those most commonly used shown on the right.

It is important to recognise that we are always communicating. Silence can imply annoyance or criticism, for example. Take time to consider and understand the effect of your personal communication style on others. Some good practices include the following.

- Control distractions as much as possible.
- Where possible, make visual and eye contact.
- Clearly identify the communicator and receiver.
- Be clear, precise and concise.
- Avoid the use of leading questions.
- Always confirm the intended message has been understood.
- Never assume meaning.
- Avoid ambiguous words, or words that could be misinterpreted.
- Encourage inclusive discussion and dialogue.
- Use phonetics for alphanumeric detail ('M for mother').
- Acknowledge communication from the listener (closed loop).

Type	Response
Closed	A fact or YES/NO.
Open	Invites an extensive reply.
Leading	Indicates the required answer.
Limiting	Restricts options.
Multiple	Many questions in one – confusion.

8.7.5 Facilitation, coaching and mentoring

Managers and team leaders should be as honest and explicit as possible about factors influencing health and safety within their workplace. This is particularly key when undergoing a change programme. Facilitation, coaching and mentoring are interventions and support mechanisms via which individuals are encouraged to take responsibility and contribute ideas to benefit safety of others and the organisation.

A facilitator is an individual who enables groups and organisations to work more effectively by collaborating to achieve results. They do not take sides or express or advocate a point of view during the meeting. They can also be a learning or dialogue guide to assist a group in thinking deeply about its assumptions, beliefs and values, and about its systemic processes and context.

Coaching is the process of enabling individuals to acquire the knowledge, skills and techniques needed to perform effectively in their occupational role by inspiring, motivating, challenging, stimulating and guiding them. Mentoring involves the long-term passing on of support, guidance and advice. It is an ongoing relationship that can last a long time and, being more informal than coaching, meetings can take place as and when individuals need guidance. Mentoring has a focus on career and personal development.

Individual commitment, ownership and accountability for safety is vital to making change happen. Everyone must be willing to accept responsibility for change in the areas they influence or control. It is important that the need to support and be supported through change is also recognised throughout the workforce, as a part of the behavioural safety culture.

Ownership of challenges is often encouraged by involving people in identifying problems and crafting solutions. It is reinforced by coaching and facilitation, incentives and, in some instances, rewards. These can be tangible (for example, financial compensation) or psychological (for example, camaraderie and a sense of shared involvement).

The most effective behavioural change programmes reinforce the core messages of safety through regular, timely engagement and communications that are both inspirational and practicable. Facilitated discussions are targeted to provide employees with the right information at the right time and to solicit their input and feedback.

8.7.6 Staffing levels

Some companies operate with the lowest possible number of people required to achieve their commercial objectives. Margins are tight, and contracts are won and lost on cost. Under these circumstances, people can be stretched beyond acceptable limits, with high workloads, long shifts and high levels of stress and fatigue. Operating without sufficient human resources can result in significant risk to the workforce.

8.7.7 Training and competency

It is likely that a workforce is more effective and efficient if it is:

- motivated and well trained
- given correct information, instruction, training and supervision
- not under unreasonable time pressure
- working with the right, well-maintained equipment.

Conversely, high workloads and tight timescales often result in training and competency assessments falling by the wayside. This can lead to ineffective decision making, poor working practices, out-of-date certification of plant, equipment and, of course, a negative effect on people's skills and behavioural safety practice.

As part of managing change, it is essential that a training and competency assessment is carried out, so that shortcomings are identified and addressed, and thus people are not put at risk. Every company is legally responsible for ensuring that people are trained and competent to carry out tasks. Greater efficiencies are achieved through correct skill levels, and further gains are made in completion times and work output.

8.7.8 Intervention

There are many recorded instances of people failing to intervene when they see an unsafe or illegal act. Whilst it is fully understandable that someone might not want to become involved in a violent confrontation in the street, in the context of work the personal risk to, say, a supervisor who intervenes to prevent someone working unsafely is generally not as great. However, the behaviour of supervisors and managers can directly affect the behaviour of operatives.

The effect of failing to intervene in an unsafe situation is to condone that activity, practice or behaviour. This in turn, sends a message to the operatives that the activity concerned is permitted, resulting in potential confusion for site teams and their working practices. Intervention by managers and supervisors is therefore crucial in every case in order to show commitment to, and support for, behavioural safety programmes. From general research and available information, the reasons for a failure to intervene appear to be split between a **lack of knowledge** that anything is wrong and a **conscious decision** not to take any action.

8.7.9 Conscious decision

The conscious decision not to intervene may possibly be based upon financial or time considerations. For example, a supervisor might ignore the unsafe use of a ladder because it saves the time and expense of hiring a mobile elevating work platform. However, there may be other personal factors for not intervening.

- Overload, where the supervisor or manager may be suffering from a heavy workload and is simply unable to identify the unsafe situation developing. Having a lack of situational awareness due to overload (for example, being unaware and failing to take notice of what is going on around them) can lead to supervisors or managers not noticing existing hazards or bad practice taking place on site, resulting in an increase in risks and accidents.
- Actions of others, especially other managers or senior managers, can shape the decisions of the supervisor. Usually the fact that no-one else involved in the operation is concerned about it is excuse enough for not getting involved. If other managers or senior managers are not concerned about unsafe actions, it may shape the decision of supervisors or managers, who may feel unwilling or uncomfortable to speak out or raise concerns. This may be the culture of the organisation – *'this is the way we do things around here'* – or may come from peer pressure or from fear of repercussions or unfair treatment for speaking out.
- Ownership of the situation where the supervisor or manager does not actually believe or understand their duties, or where they are not directly in charge of the operation and believe they have no jurisdiction. If a supervisor or manager feels they have no clarity about taking ownership of safety issues, or no authority to intervene, it may be because they fear lack of support from others or fear being overruled or marginalised for the intervention. They may feel they should not get involved and, instead, adopt a 'not my job' mentality, believing it is not their problem and someone else should deal with it.
- Where a supervisor or manager lacks knowledge about the task or lacks the important communication skills, then they are less likely to get involved – having the skills to resolve the issue is important. If a supervisor or manager is not suitably knowledgeable about a task, or doesn't have the right communication skills, they may avoid taking responsibility and not get involved. They may feel embarrassed by a lack of knowledge, unable to effectively intervene and communicate what is required, and fear that they will be criticised if they get it wrong. Supervisors and managers should be given training in communication skills and be empowered to act if they feel a situation is becoming, or is, unsafe.
- The risk of possibly entering into a situation where they may be required to make a difficult decision that could have a significant effect on the project. The support of senior managers is critical to allow junior managers and supervisors to become involved in safety issues and empower them to take whatever action they deem necessary if an unsafe situation arises. This could even involve the cessation of work until the safety issue is investigated further.

8.7.10 Managing the risk

Making assessments about risks and reaching an informed decision requires access to the information held within the safety management system. The process of obtaining information begins with the recognition that the problem exists, and then raises questions to which answers are required.

Deciding on the level of accuracy and precision of the risk and safety assessment will depend on the sampling and measurement methods used in collating the initial risk information. In behavioural terms, this is done through identifying 'what' is happening during an observation and asking 'why'. The 'whats and whys' are analysed and tabulated to identify trends, often by interpreting the data.

Interpretation is based on the personal perception of what has been observed and so identifying trends can be difficult. Although risk can be quantified as abstract principles, health and safety cannot.

Whilst risk assessment is based on knowledge of the job and past experience, the corresponding judgement on safety is subjective and can be political. It may be possible to obtain group agreement on objective and rational measures of risk for various work. However, there will often be controversy over what are considered to be safe conditions. Attempting to define acceptable levels of risk immediately raises the question of 'to whom' or 'on what terms' is the risk acceptable?

The distinction between risk and safety is relevant to many organisations and there are a number of factors to be considered in defining the acceptability of risk (see *overleaf*).

BEHAVIOURAL SAFETY

Cost. Safety is always compromised by available budget yet it costs far more to investigate and restore safe working conditions after an accident than it does to resolve the issues in the first place.

Controls. Who has control? Those at the place of work should have control over the safety requirements of the task. Ownership of health and safety controls is critical for a safe working environment.

Customs. Many risks are taken because certain work has always been done that way.

Conditions. Many people are put at risk because conditions have changed, resulting in longer working hours, tight timescales, lack of resources, workload, fatigue, stress or an ageing workforce. This can lead to an increase in overall errors, and to more serious errors occurring.

Consequences. Managers rarely evaluate in advance the consequences of something going wrong. Often the thought process seems to be 'if it hasn't happened yet, it won't happen at all'.

Benefit. What benefits does the individual get from taking a short cut, such as an early finish when the job is done?

8.7.11 Benefits of health and safety discussions

One method of enhancing any safe system of work is through frequent and open discussions. The heart of any process is communication, with all involved recognising the value of shared ideas and knowledge. This has the potential to impact positively on bottom-line profits, with individuals and teams working more efficiently towards achieving high-quality business objectives. The company's image will also benefit if, by the actions exhibited, it is shown to be committed to a safe and healthy working environment, where no-one is injured or becomes ill as a result of coming to work.



An example of good-practice communication is for site managers to have an informal 10-minute chat with their employees and/or contractors' supervisors at the start of every day. The manager should encourage them to tell each other where they will be working and how their activity could affect other people. This will help supervisors to plan their day, as well as improve co-ordination, consultation, production and, ultimately, safety.

The aim of a behavioural safety discussion is to identify any difficulties in completing tasks safely and to aid the supervisor, manager and individual in identifying problems and achieve a safe system of work. Participants in the discussion should be encouraged to use open questions, smooth the way forward, be clear in what they are saying, avoid any misunderstanding and proactively resolve issues through positive actions rather than reactively observing unsafe actions.

Those with more knowledge and experience can assist newer colleagues in understanding the hazards around them and stop people putting themselves at risk. Learning through a coaching and mentoring approach can be more effective and less onerous than more formal observation work.

All workers and managers can make valid contributions to discussions on behavioural safety. Do not impose your own thoughts and avoid passing judgement. Above all else, problems or issues should be resolved immediately with someone who has the authority to make the necessary changes.

CONTENTS

Leadership and worker engagement



9.1	Introduction	120
9.2	Understanding worker engagement	121
9.3	Tools and techniques, hints and tips	121
9.4	National Occupational Standards – Management and leadership	122
9.5	Legislative requirements	122



GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

LEADERSHIP AND WORKER ENGAGEMENT

Overview

This chapter gives a general overview of the principles of leadership and worker engagement. It looks at how these principles affect organisations within the construction industry. Additional resource information, tools and techniques are presented, which can inform workplaces and influence leadership and engagement behaviours.

9.1 Introduction

The construction industry is one of the UK's most significant economic sectors, with over 353,365 registered construction companies operating in the construction industry across Great Britain. Construction is also consistently one of the most dangerous sectors to work in, according to Health and Safety Executive (HSE) statistics.

The industry is therefore continually seeking to find innovative ways to reduce ill health and accidents by engaging with its workforce. This has a number of identifiable business benefits, including the promotion of sustainable behavioural change to ensure that health and safety is taken seriously on site.

Success depends upon the quality of leadership and management practice within organisations. Commitment and responsibility at all levels are critical to the development of a successful, innovative business and proactive worker engagement.

Definitions of leadership vary across organisations and leadership experts. Leadership is largely centred around building relationships and interpersonal skills, focusing on individuals who are:

- visionary
- motivational
- inspirational
- emotionally intelligent
- trustworthy
- natural leaders and communicators
- driven
- ambitious
- resilient – able to deal with pressure and failure.

There is a need for a range of skills and knowledge. These fall into three main areas.

1. Appropriate technical and professional skills, relevant to the organisation.
2. Commercial and financial, including a high level of business acumen and experience (both success and failure).
3. Skills in people management and development (coaching and feedback, communication and team management).

Good leadership is seen as encompassing a strong mix of all these skills, with weaknesses in one area being seen to diminish the overall capability of the leader. Research has identified various traits, differentiating good from outstanding leadership. These outstanding leaders are described as having the following abilities.

- Being able to think and act systematically.
- Understanding the whole and the connection between internal work and the external environment.
- Seeing people as the route to performance through high performing teams.
- Building relationships and providing an environment in which teams and people can shine.
- Being self-confident without being arrogant.
- Being professional.
- Having personal humility.
- Providing appropriate space and the support to allow people to think and make decisions.

A vital task facing leaders is to engage workers within the day-to-day operation of their own roles in order to achieve the wider organisational objectives. Leaders who are able to bring the full range of worker motivation, creativity, skills and knowledge to their businesses are those whose organisations succeed in an increasingly competitive and fast-moving market place.



Leadership development, worker engagement and business performance are strongly linked. In organisations where management and leadership development are embedded, employee engagement levels are higher.

9.2 Understanding worker engagement

9.2.1 Work Foundation

The Work Foundation, a provider of research-based analysis, knowledge exchange and policy advice in the UK, defines employee (worker) engagement in the following way.



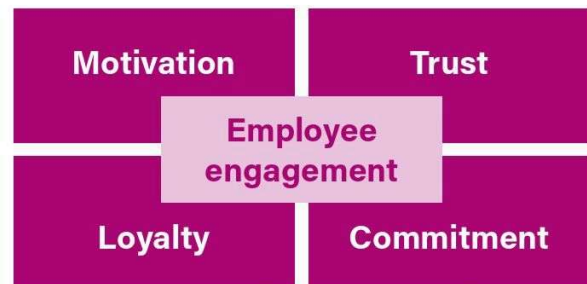
Employee engagement describes employees' emotional and intellectual commitment to their organisation and its success. Engaged employees experience a compelling purpose and meaning in their work and give of their discrete effort to advance the organisation's objectives.

Effectively, engagement is about how workers behave at work and the impact this has on their own performance and that of the organisation, in particular with respect to the discretionary effort they apply within the workplace.

Within the construction industry, there is a specific focus on health and safety within worker engagement practice.



For further information visit the Work Foundation website.



A typical representation of factors influenced by employee (worker) engagement strategies

9.2.2 Barriers to workforce engagement

Research by the HSE has identified that barriers to workforce engagement are, in many ways, similar to behavioural change barriers.

- The disparate, transient and multicultural nature of the workforce on site.
- Working style associated with a results-focused and target-driven industry.
- Disincentives to report health and safety issues that may result in future loss of projects and income.
- Lack of client motivation to adopt worker engagement practices.
- Appropriateness of off-the-shelf solutions, given the variance in industry-related organisations.
- Lack of management commitment to, or understanding of, the benefits of worker engagement practice.

The following list shows some potential solutions to **overcoming barriers**.

- Treating each engagement project as a separate strategy.
- Training relevant individuals in behavioural change and worker engagement skills.
- Focusing on the supply chain as a route for delivery of worker engagement.
- Including the supply chain and sub-contractors in behavioural change and engagement training.
- Running launch events for worker engagement and building behavioural change and worker engagement in to sub-contracts.
- Building a preferred supply chain.

In addition, there is a real focus on worker engagement tools that can be employed to aid implementation of good practice. This is particularly the case for the construction industry, where a lack of worker engagement is directly linked to the potential for health and safety incidents.

9.3 Tools and techniques, hints and tips

9.3.1 Engaging workers

- The process of engaging workers takes time; those that have done it say that while results were noticed within a year, it was nearer five years before the process was embedded. The process is **never** complete.
- Make sure the directors and senior managers visibly support a worker involvement culture. They can do this by addressing meetings, sending out messages, instructing managers, and 'talking the talk and walking the walk'.
- Explain why you want to talk to workers and what it involves.
- Run an employee opinion survey and be seen to act rapidly on suggestions or shortcomings. Employees will then start to appreciate that the employer is serious about worker engagement.
- When receiving suggestions, ensure that the person making the suggestion receives feedback, even if the answer is 'sorry, we can't, and these are the reasons why'. The fact that the views are treated seriously is usually more important than agreeing with them. Ensure that you publicise responses to employees' suggestions.

LEADERSHIP AND WORKER ENGAGEMENT

- Talk to personnel at all levels, using different tactics for different groups and even individuals. For example, email might elicit suggestions from office-based workers, but may be less appropriate on a construction site.
- Be visible. Walk around. Talk to staff. Take small numbers of staff on regular safety walk rounds.
- An anonymised system of reporting complaints or problems is helpful in the initial stages of embracing worker engagement, but appears to be rarely used once the culture is embedded.
- Ensure you take the views of shift-workers and part-time staff into account.
- Take staff to another organisation where worker engagement is working.
- Worker engagement in health and safety will not happen without a genuine no-blame culture.
- Ensure representatives have suitable training for the role (covering eliciting views, presenting a case, feeding back to colleagues) as well as in health and safety. Even in the best examples, representatives are often not trained in this role.
- As people are often reluctant to volunteer as a representative of employee safety, it may be worth speaking with employees you think would do a good job. Assertive and confident individuals facilitate better engagement than 'yes' people. Informal leaders (who workers already turn to for advice) may make good representatives themselves, or be important in lending credibility to the notion and practice of engagement. Consider also offering incentives or rewards (financial or otherwise).
- Make sure that joint health and safety committees have a fair balance of employee representatives and managers. Managers may consider absenting themselves from part of a meeting in order to ensure representatives are not intimidated from speaking. This should not undermine the case for strong leadership from the top, and the need for such absence will lessen as representatives grow in experience and confidence.
- Ensure your managers and staff are trained in soft skills as well as hard skills.



Worker involvement in health and safety will not happen unless there is a no-blame culture in place.



There is no doubt that the quality and effectiveness of management and leadership has a significant impact on the level to which workers are engaged within their jobs. This, in turn, has direct consequences for health and safety measures within organisations.

9.4 National Occupational Standards – Management and leadership

The National Occupational Standards (NOS) for management and leadership are statements of good practice that outline the performance criteria, related skills, knowledge and understanding required to effectively carry out various management and leadership functions. They are produced by the Management Standards Centre (MSC), the Government-recognised standards-setting body for the management and leadership areas. The MSC describes the standards as:



the activities/functions of management and leadership at various levels of responsibility and complexity. Therefore, they are relevant to anyone for whom management and leadership is, to a greater or lesser extent, part of their work. This applies to managers and leaders in all sizes and types of organisation, and in all industries and sectors.

This information presents the suite of standards most relevant to different levels of management and leadership. These standards are generic and so equally applicable to management and leadership practice within organisations of all sizes and across all sectors.



For further information visit the NOS section of the Government's website.

9.5 Legislative requirements

The Health and Safety at Work etc. Act 2(4) allows regulations to be made for appointing safety representatives from recognised trade unions.

As a result the **Safety Representatives and Safety Committees Regulations 1977** were introduced. These were later amended by the Management of Health and Safety at Work Regulations.

The **Health and Safety (Consultation with Employees) Regulations 1996** widened the scope of the duties of employers to consult with employees who are not members of a trade union.

Additionally, the **Construction (Design and Management) Regulations 2015** (CDM) have placed an increased emphasis on consultation. There is also a specific duty on the principal contractor to consult and engage with workers.



Where an employee is a member of a recognised trade union, the Safety Representatives and Safety Committees Regulations 1977 will apply, together with the Health and Safety at Work etc. Act 1974. Where there is no recognised trade union membership representation, the Health and Safety (Consultation with Employees) Regulations 1996 apply.

9.5.1 Legal requirements for communicating with all employees

Employers must inform employees of the following points.

- The main terms and conditions of employment.
- Changes in terms and conditions of employment.
- The reason for dismissing them from their job, where applicable.
- Certain business matters (for example, when there is a significant change in the way the business operates).
- Health and safety issues and consultation with the workforce on health and safety.

The Health and Safety (Consultation with Employees) Regulations and the Safety Representatives and Safety Committees Regulations will apply to most workplaces.



The HSE has developed easy-to-use guidance that shows the relationship between the two sets of regulations and how they may affect businesses and their workforces.

The business may benefit from consulting employees on a regular basis and making staff aware of ways they can contribute ideas and raise concerns. The business does not need to have complex structures for consultation – often ad hoc groups can work best.

For effective consultation consider the following.

- Explaining final decisions, particularly when employees' views are rejected.
- Giving credit and recognition to those who provide information that improves a decision.
- Ensuring that the issues for consultation are relevant to the group of employees discussing them.
- Avoiding minor issues and petty grievances.
- Making the outcome of the meeting available to everyone.

Good communication processes will make employee engagement easier and more effective. The main aspects of achieving that include the following.

- Involving the workforce in building a set of values by which the company will operate.
- Building dialogue with the workforce by communicating regularly and openly about matters that affect the business. This will build credibility and create an environment of trust.
- Involving the workforce in the most important employee issues as soon as possible.
- Avoiding leaks of information or rumours spreading.
- Living the values.

In the construction industry it is especially important to ensure that this communication is effective and timely, particularly in relation to health and safety. All workers have a right to be consulted on the work they do, to be given adequate information and training to carry out this work in a safe and healthy manner, and to have fit-for-purpose site facilities. Being directly involved in the process, employees often see problems as they emerge.

Recent years have seen a large increase in the number of foreign workers in all sectors. English may not be their first language and communication can become more difficult, leading to a greater risk of misunderstanding. These workers must be fully engaged and consulted on matters of health and safety at work, and employers must find adequate mechanisms to do this. Similar consideration should be given to other workers who have difficulties in communicating.

CDM place a duty on principal contractors to make and maintain arrangements to enable effective co-operation between all parties on site, and to consult with all workers on site. Consultation means not only giving information to workers, but also listening and taking account of what workers say, before making health and safety decisions. Part of the purpose of consultation is to make sure the measures taken on site to protect workers' health and safety are effective. Principal contractors are encouraged to develop a variety of methods of communication and consultation with the workforce to develop collaboration and trust. When matters of concern are raised by workers these should be taken seriously and feedback given.

Evidence that this is happening provides assurance that effective worker engagement is in place. Involving the workforce in identifying and controlling risks is crucial in preventing accidents. Whether projects are notifiable or not, contractors have a duty to inform workers of their procedures for stopping work in the event of serious and imminent danger, and to provide training where necessary. This should make sure workers are willing and able to intervene to prevent an accident from happening.

9.5.2 Safety Representatives and Safety Committees Regulations 1977

A summary of the **role of the safety representative** is shown below.

- Representing their members' interests in matters of health, safety and welfare.
- Representing their members in consultation with the HSE or other enforcing authority.
- Carrying out the statutory functions outlined in the regulations.

LEADERSHIP AND WORKER ENGAGEMENT

Functions of the safety representative include the following.

- Consulting with the employer.
- Investigating and reporting significant hazards and dangerous occurrences.
- Investigating accidents in the workplace.
- Providing representation on general health and safety matters.
- Receiving complaints by employees.
- Providing representation in consultation with the HSE or other enforcing authority.
- Receiving information from the enforcing authority.
- Attending safety committee meetings.
- Inspecting the workplace.



For further information on consulting workers on health and safety visit the HSE website.

9.5.3 The Equality Act 2010

Equality and diversity are issues of rising importance that the construction industry needs to acknowledge and address. This has been brought into sharper focus following the recent changes to legislation, with the introduction of the Equality Act, which became law in October 2010. It replaces previous legislation (such as the Race Relations Act and the Disability Discrimination Act) and ensures consistency in what employers need to do to make the workplace a fair environment and to comply with the law.

The Equality Act covers the same groups that were protected by existing equality legislation – age, disability, gender reassignment, race, religion or belief, sex, sexual orientation, marriage and civil partnership, pregnancy and maternity – but extends some protections to groups not previously covered, and also strengthens particular aspects of equality law. It is a mixture of rights and responsibilities that have:

- stayed the same – for example, direct discrimination still occurs when someone is treated less favourably than another person because of a protected characteristic
- changed – for example, employees can now complain of harassment even if it is not directed at them, if they can demonstrate that it creates an offensive environment for them
- been extended – for example, associative discrimination (direct discrimination against someone because they associate with another person who possesses a protected characteristic) covers age, disability, gender reassignment and sex, as well as race, religion and belief and sexual orientation
- been introduced for the first time – for example, the concept of discrimination arising from disability, which occurs if a person with a disability is treated unfavourably because of something arising as a consequence of their disability.

Whilst construction sites can be hazardous places of work for anyone, employers must accept their legal duties and make reasonable adjustments within the working environment to ensure that fairness is achieved and discrimination does not occur.



Impacts of equality and diversity on the health and safety environment

Equality and diversity impact on the health and safety environment because:

- staff who are bullied or suffer harassment at work may not feel safe or act in a rational manner – bullying/harassment, including initiation ceremonies and horseplay, banter and fooling around can also involve unsafe acts
- a person may have a visual, hearing or speech impairment that could, for example, make communication of risks difficult
- a person may have restricted mobility and therefore have difficulty in walking, climbing ladders, lifting or carrying out certain aspects of work
- poorly fitting personal protective equipment (PPE) for employees can be uncomfortable, impede mobility and/or movement necessary for safe working (for example, a person's physique may mean standard-issue PPE is too small or large).

A part of making reasonable adjustments is likely to be consultation between the employer and the person(s) on how certain situations will be managed. Employers have a responsibility to ensure their employees are supported and not discriminated against. Supervisors and managers must be aware of these responsibilities and the way to eliminate or effectively manage situations if they arise.

In addition, employers have a responsibility to ensure every employee understands safety information and messages, both in terms of language used and the workplace culture. Ensuring clear messaging can result in better working conditions and practices and a more productive team approach.

If women are experiencing menopause symptoms that have a long-term and substantial impact on their ability to carry out normal day-to-day activities, these symptoms could be considered a disability. If symptoms amount to a disability, an employer will be under a legal obligation to make reasonable adjustments. They will also be under a legal obligation to not directly or indirectly discriminate because of the disability, or subject the woman to discrimination arising from disability.

Women experiencing menopause symptoms may also be protected from direct and indirect discrimination, as well as harassment and victimisation, on the grounds of age and sex. Employers have a legal obligation to conduct an assessment of their workplace risks.

CONTENTS

Inspections and audits

10

10.1	Introduction	126
10.2	Important points	126
10.3	Measuring performance	126
10.4	Monitoring	127
10.5	Inspections	127
10.6	Hazard-spotting techniques	128
10.7	Inspection reports	129
10.8	Audit and review	129
10.9	Remedial actions	130
10.10	Benchmarking	131
10.11	Additional benefits to a company	131

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

INSPECTIONS AND AUDITS

Overview

Measurement is a vital step in the management process and forms the basis of continual improvements. If measures (such as site inspections or health and safety audits) are not carried out correctly, there is no reliable information to show if health and safety risks are being managed.

This chapter outlines the types of measurement and the benefits of monitoring health and safety.

10.1 Introduction

Employers and employees have a moral, social and legal duty to prevent accidents by every practicable means available to them.

The costs of getting it wrong, to a company, its employees and others affected, may be great and provide ample motivation to all involved to reduce the risks to health and safety.

Motivation must, however, be translated into effective action, and this can only be achieved by the commitment of management and supervisors at all levels. It is well established and documented that accidents can be prevented by:

- identifying the hazards that employees face within the workplace
- understanding how accidents are caused by unsafe acts, unsafe systems of work and unsafe conditions on site
- taking steps to control the activity of the worker, the work method and the workplace.

Employers, managers, supervisors and safety representatives all have equally important roles to play.

By obtaining and providing information through the **inspection, investigation and examination** of the workplace, they can help provide a basis for effective management action to promote safer and healthier workplaces, and induce a greater awareness of health, safety and welfare on the part of all concerned.

There are strong links between this topic content and the need to speak with employees on matters of health and safety.

10.2 Important points

- Health and safety inspections and audits can become a productive part of consultation between management and the workforce.
- They are often carried out by directors, managers or health and safety professionals, and are a key element of the site management teams role.
- The successful outcome of any inspection or audit is that remedial actions are put in place where shortcomings have been identified.
- Workplace inspections are most effective when carried out against a predetermined checklist incorporating some method of recording the findings.
- Whereas workplace inspections tend to be a 'snapshot in time', an audit is a thorough examination not only of the site conditions prevailing at any one time but also of the:
 - commitment of management to health and safety
 - procedures and systems that underpin the health and safety management system.
- The emphasis on contractors being able to demonstrate their competence in matters of health and safety management potentially puts a greater importance on them being able to show that inspections and audits are carried out and acted upon.

10.3 Measuring performance

There are several ways in which a company may measure the effectiveness of its health and safety performance.

- Evaluating the effectiveness of earlier risk assessments.
- Gathering and evaluating information on the number of accidents over a reference period, their causes and effects.
- Evaluating worker feedback on near misses and dangerous occurrences.
- Gathering information on reportable diseases.
- The results of health and safety audits.
- The results of inspections, whether regular, unannounced or specific.
- The level of health and safety awareness among employees in the form of feedback from training courses and safety committee meetings.
- Benchmarking against previous audits.

10.4 Monitoring

Employers are required to monitor performance. There are two types of monitoring used in auditing and inspecting: active and reactive.

10.4.1 Active monitoring

Active monitoring (often referred to as *proactive monitoring*) measures current levels of compliance with legislation and company procedures. Effective active monitoring will help to reduce accidents and ill health in the longer term, and can take the following forms.

- Health and safety inspections
- Pre-use checks
- Safety tours
- Hazard-spotting exercises
- Health surveillance
- Audits.

Health and safety inspections tend to be the site manager's snapshot of the health and safety standards on site at any one time. Alternatively, or in addition, a company's health and safety adviser or director may carry out these inspections. A health and safety inspection or audit may examine the big picture or concentrate on only one feature. Irrespective of who carries out an inspection, they must have the knowledge to detect unsafe situations and the authority to ensure these are rectified.

10.4.2 Reactive monitoring

This type of monitoring investigates anything that has actually happened. Accidents, incidents, near misses and illness resulting in serious injuries and fatalities, including health hazards, are too often a feature of work in the construction industry. The active management of health and safety should serve to keep such incidents to a minimum.



It should not be forgotten that a near miss today could be a potential accident tomorrow. Everyone has an important role to play in both active and reactive monitoring.

10.5 Inspections

Formal inspections at reasonably regular intervals should add to site managers' day-to-day checks, inspections and examinations, which occur as part of any task. These should follow a properly designed checklist for the systematic inspection of the workplace.

Advantages of regular inspections include that they ensure good housekeeping is maintained within the workplace and that awareness is developed, amongst employees at all levels, of the need to promote and maintain safety, health and environmental standards. The disadvantage or danger of regular inspections is that they may become a rather mechanical routine for all concerned, and that their impact might be lessened.

Carrying out joint inspections with safety representatives, other members of the team or even nominated workers from different trades or contractors can help to identify issues that may be overlooked during a routine inspection.

Feedback and results can be jointly presented to the workforce via various communication methods in place, such as toolbox talks. Stopping and engaging in a two-way conversation with workers during inspections is vital, to understand their way of working and any issues and concerns.

This goes a long way in helping to build a positive safety culture, but it is just as important to follow through or feed back on issues. More often than not, workers will be able to help find and agree solutions. Random inspections carried out without any prior notice to the workforce, on different days of the week, at irregular intervals and at different times of the day, avoid the shortcomings of a predictable inspection and help to encourage a continuous interest in health, safety and environmental matters by all personnel. In practice, a combination of both regular and random inspections is encouraged.



Inspection apps for smartphones and tablets enable instant emailing of pictures and reports

10.5.1 Safety representatives

In accordance with the **Safety Representatives and Safety Committees Regulations 1977** safety representatives are entitled to inspect the workplace. An Approved Code of Practice (ACoP) and guidance notes have been issued by the HSE to support the relevant regulations. These emphasise areas where safety representatives should reach agreement with their employers.

A safety representative may consider the company's safety inspection procedure to be adequate. In such cases, the representative may only need to seek an agreement with the management that they are involved in any future inspections. However, if the company's arrangements for the inspection are thought to be inadequate, recommendations should be made in writing in order that action may be taken to remedy the situation.

INSPECTIONS AND AUDITS

10.6 Hazard-spotting techniques

Hazard-spotting exercises are a modified version of the safety sampling method, where a particular department, hazard or work activity is singled out for a closer and more thorough examination.

The exercises should be arranged via collaboration between management and the health and safety team, or by the training manager through line management, and should involve supervisors, safety representatives and operatives. Observers may include first line or senior managers, but there should be at least an equal number of operatives. Some managers may prefer a predominant number of operatives up to foreperson level in the team, thereby encouraging workers to take an active interest in health and safety and accident prevention in the workplace.

10.6.1 Hazard-spotting programme

A properly structured programme of inspections and sampling will ensure that all of the main activities of a company are continuously under scrutiny. Hazard-spotting exercises should be used not only where there is seen to be a need to ensure the safety and health of people at work, but also as a preventative and continuing monitoring exercise.

10.6.2 Hazard-spotting team

Members of the proposed hazard-spotting team should have had some training or experience in health and safety matters in relation to the work or hazards to be assessed and be able to recognise any unsafe acts of people at work or unsafe or potentially unsafe conditions of work or methods of work. One person should be appointed leader of each group, and their functions should include the following.

- The collection of the written findings of the members of the team.
- The study of these findings and the summary of unsafe acts and conditions that have been observed during the tour.
- The preparation of a brief report, setting out the unsafe acts and conditions that were observed during the tour and making recommendations for action by the management or their representative to rectify any situation that may have been observed.
- Noting and recording the response of the line or senior management, satisfying themselves that action will be taken by management to rectify any situations observed.
- Reporting back to members of the team what action will be taken by the line manager concerned and when the action will be taken.

10.6.3 Safety sampling

Health and safety sampling of a particular work activity, process or work area may be necessary in the following situations.

- The activity, process or work area presents particular health and safety concerns, where there have been changes to an activity, process or work area that are relevant to health and safety.
- There is a need to improve the health and safety performance in a particular area of operations.
- There are areas of high labour turnover.

Sampling should be carried out by someone familiar with the work activity, process or work area under inspection.

10.6.4 Safety tours

A variety of roles, such as directors, senior managers, site managers, engineers, supervisors and operatives, should have the opportunity to observe health and safety conditions prevailing within the workplace during normal construction work.

Production, however, will often be their first priority and this need to get the job done may adversely affect their judgement on health and safety matters. Familiarity with certain work and hazards may further cause them to overlook, fail to recognise or ignore real or potential dangers that are present at the workplace.

For these reasons, additional **health and safety tours** (general inspections of the workplace) made by competent health and safety professionals, supervisors and safety representatives should take place at regular intervals.



A site health and safety tour



For HSE guidance aimed at directors and board members, covering the identification and management of health and safety risks at work, visit the HSE website.

10.6.5 Health and safety surveys

These are carried out at longer intervals to assess current health and safety standards and activities.

They can be compared against other departments in the same organisation or similar companies within the industry.

Surveys provide the following opportunities.

- Compare the current health and safety performance of the company against previous years.
- Review the understanding of health and safety objectives and determine future objectives and key performance indicators (KPIs).
- Establish adequate and suitable current levels of health and safety training.
- Improve procedures, records, communication and information.
- Decide if additional resources are required.

Surveys can be anonymous to encourage an open and honest response and they can also encourage worker engagement.

10.7 Inspection reports

Details of inspections and their findings should be recorded. Depending on the company, inspection checklists can be recorded and shared via electronic devices, or by using paper versions.

Whatever the type of checklist or form used, it should provide a record of any action requested to remedy conditions and working practices considered to be unsafe. The date and place of the inspection and details of any corrective action identified, subsequent action to be taken or, if no action was required, an explanation of why this was so should also be recorded.

Reports should also contain elements of positive or good practice, such as identifying that safe systems of work for high-risk activities were being followed. Good practice should also be captured and shared and opportunities made to offer recognition to individuals who are working safely.

Remedial actions identified to eliminate or lessen the risk to health and safety should always be implemented (*refer to 10.9*).



For further information on inspections and reports visit the HSE website.

10.8 Audit and review

A health and safety audit is a demonstration of the management's commitment to monitor and improve, where necessary, the effectiveness of the company's health and safety management system from the top down.

The process of carrying out health and safety audits, evaluating the findings and implementing remedial action, where necessary, results in a continuous cycle of improvement.

Audits tend to be formal, pre-planned, events with the findings documented for later evaluation and review.



Audit

An **audit** is the structured process of collecting independent information on the efficiency, effectiveness and reliability of the total health and safety management system and identifying plans for corrective action.

Therefore, the main requirements of a health and safety audit include the following.

- A critical examination of the whole operation.
- An assessment of how the risks to health and safety are being managed.
- The identification of the efficient and safe performance of people.
- The detection and identification of falling standards, ineffective company procedures, and non-compliance with industry standards and legal requirements.
- The use of meaningful standards consistent with the organisation's operations.
- The identification and implementation of a corrective action plan.

Health and safety auditing should also be seen as an integral part of the overall monitoring of health and safety within an organisation.



The aim of a health and safety audit is to identify problem areas that may exist, so that you can make improvements to your standard of health and safety. The audit should look at the interaction of all the activities of your company, as well as at the work itself.

Audits may discover health and safety failings that arise through factors that occur off site (such as design issues), which are therefore outside the direct control of site-based staff. They may also identify good practices that should be shared and incorporated into a process of improvement. By comparison, health and safety inspections are normally a less formal style of audit; the results are not necessarily recorded and they may be more appropriate for some smaller sites. Remember, however, **no record – no proof!**

INSPECTIONS AND AUDITS

Both health and safety audits and inspections are carried out on a pre-notified or no-notice basis, as suited to the situation at the time. If too many are pre-notified or regularly scheduled, it may tend to create a false impression, as those on site may only improve their behaviour or practices in the short-term.

Audits should be carried out by a member of management, often the company health and safety professional or a director, who may or may not be accompanied by a representative(s) of sub-contractors or the employees.

The aim is to identify problem areas, implement improvements and increase the overall standard of health and safety awareness.

It requires attention to the following.

- Work environment.
- Person carrying out the task.
- Task itself.
- Safety culture.
- Way in which each affects the others.

The important thing is that, where shortcomings are found, remedial action is promptly taken and lessons learned to prevent recurrence.

10.9 Remedial actions

A clear understanding of the remedial actions required to eliminate or lessen the risks to health and safety, and when those actions should be taken, is an essential part of health and safety management.

The following lists suggest how shortcomings that are highlighted during health and safety inspections should be prioritised for remedial action.

10.9.1 Items requiring immediate action

- Any accident or incident that is reportable to the HSE under the Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013.
- The occurrence of accidents, incidents or near misses that produce situations of possible or immediate danger to the health and safety of employees or other people, including members of the public.
- Any activity or situation that is giving rise to conditions that could affect the short or long-term health of workers (for example, creating dust).
- The contravention of any statutory requirements or ACoP.
- The risk of financial liabilities, as a result of damage to plant and equipment, or compensation to workers or members of the public following litigation, the risks of fire, explosion or other hazards involving electricity, toxic materials or substances.
- The existence of unsafe working practices and unsafe places of work.
- Any shortages of correct and adequate personal protective clothing and equipment.
- The risk of environmental pollution.

10.9.2 Items requiring prompt action

- Any potential hazards that may exist, but which do not cause any imminent or immediate danger of injury, damage or pollution.
- Any signs of inadequate information, instruction or supervision, which should have been provided by either the management or others.
- Any first-aid facilities and training that fall short of statutory requirements.
- Occasions when new plant, new work methods, new equipment or different materials are to be introduced into the workplace.

10.9.3 Items requiring short-term action

- Where there is a lack of planning and control affecting safety, occupational health or the environment within the workplace, either directly or indirectly (for example, through the inadequate supply of materials and equipment to enable the workforce to carry out their tasks satisfactorily and safely).
- Where there are signs of inadequacies in the personal skills, knowledge and experience of the workforce, which may have an adverse effect on safety.

10.9.4 Items requiring long-term planning and action

- Where there is a lack of certain categories of safety skills and trained personnel amongst the workforce.
- Where there is a need for the training of safety advisers, supervisors and safety representatives, to keep abreast of the future needs of the company and its employees.
- Where there is a need for the improvement of standards of health, safety and welfare within the company.

10.10 Benchmarking

One of the most difficult things about auditing is deciding whether the measured performance is satisfactory. Within health and safety, experienced external auditors will typically audit against two parameters.

- Performance against internal systems.
- Performance against legal standards.

This is acceptable as it allows an organisation to self-check, but it does not indicate how a performance rates against other companies.

As a comparative measure, some companies use accident or injury numbers, or the number of prosecutions and/or enforcement notices.

Various attempts have been made within the industry to produce key performance indicators (KPIs) for health and safety, but difficulties are caused by comparing the relevance of, say, KPIs of a small plastering contractor employing four people to a Build UK member.

As a result there is no right or wrong answer, and it is a case of selecting areas of measurement most appropriate to a company. Another driver for benchmarking certain elements is that these may be a requirement of client or pre-qualification questionnaires.

Some trade associations, including the National Access and Scaffolding Confederation (NASC), compile accident statistics for their members. NASC believes that their members represent approximately half of the scaffolding industry in the UK; as a result, the accident statistics gathered may not be a true representation of accidents across the whole scaffolding industry.

In recent years active indicators are measured in addition to the traditionally reactive indicators (such as accident statistics). Active indicators can include the number of:

- toolbox talks carried out per employee
- senior management tours
- site inspections and audits
- near-miss reports.

Benchmarking can be a useful exercise but participants need to be aware that the information may need to be interpreted carefully.

Health and safety performance can be assessed in the following way.

- Relating it to the same reporting period of the previous year.
- Comparing it with similar organisations and/or departments in the same industry.
- Relating it to official national statistics.
- Making regional comparisons between sites in different parts of the country.

In a market of open communication between managers, such comparisons should be extremely useful and readily available to help identify and address trends, and take appropriate action where common hazards have been recognised.

10.11 Additional benefits to a company

As legislation increasingly requires risk assessments and other assessments, clients, principal contractors and contractors are demanding to see health and safety policies and method statements. Evidence that you have an auditing programme will help your business.

A good audit will benefit everyone from senior management to the most inexperienced employee, and should be accepted on its merits as the resultant changes bring improvements in health and safety performance.

The health and safety audit should be carried out, as far as possible, by independent auditors. This overcomes the problems faced by line managers when auditing their own area of work, or even a second line manager being critical of their peers. Auditing must be carried out objectively and with a high degree of honesty when identifying non-compliance with management systems and techniques.

The frequency of audits should ensure that the health and safety management system of the organisation does not degrade over time or through changes in the company's organisation, personnel or the work it carries out.

Several health and safety auditing systems are available commercially but, in the main, these are directed towards the larger business, which can afford either to employ its own dedicated auditing team or contract consultants to come in and carry out the audit.

The construction industry identified the need for an auditing system for the smaller and medium-sized company where the site manager or site supervisor could inspect the site and, within a couple of hours, produce an acceptable appreciation of the standard of safety on site.



CONTENTS

Statutory forms, notices and registers



11.1	Introduction	134
11.2	Important points	134
11.3	Record keeping	134
11.4	Daily user and visual checks	135
11.5	List of key requirements	135
11.6	Statutory or recommended inspections and examinations	137
	Appendix A – Brief explanation of key requirements	138



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

STATUTORY FORMS, NOTICES AND REGISTERS

Overview

This chapter gives a brief outline of some of the requirements for the completion and use of various statutory and non-statutory forms, notices, signs and registers used within the building and construction industry, and the keeping of records and other details.

It also includes guidance on the types of daily and weekly inspection work that should be carried out and records that must be kept.

11.1 Introduction

Record keeping is important as it is the means to ensure that inspections and examinations are carried out. It ensures that there is a history of what occurred at what point during a project, and can be referred to if necessary, for example if needing to defend and protect a company against a claim.

Some records are required by legislation and others by the employer as they are good practice and vital in preventing accidents.

11.2 Important points

- The selection of forms or notices should be appropriate or applicable to the individual site or premises, and the circumstances that exist on that site.
- The listings contained in this chapter are not intended to be a full and precise guide to all the statutory requirements that currently exist. All cases of doubt should be resolved by reference to the appropriate enforcing authority or to the legislation.



A statutory inspection means legally it must be carried out.

11.3 Record keeping

Records are generally split into two categories.

Active (proactive) (for example, inspecting scaffolding or excavations to ensure they remain safe).

Reactive (for example, an accident investigation to establish the causes).

Records of toolbox talks are also important, as this demonstrates that (usually site-based) and on-going awareness training is being provided.

Daily site briefings are a good way of reviewing what has happened and looking at what has to be done and the associated hazards for the forthcoming day. These can be extended to discussing point of work risk assessments.

Health surveillance records must be kept for all employees under health surveillance. They must be kept for at least the period specified in the relevant regulations, from the date of last entry. For example, this would be 40 years under the Control of Substances Hazardous to Health Regulations 2002.

Where regulations do not specify how long they should be kept for, the health record should be kept at least while the worker is employed.

Inspection records must be kept, and the duration will be specified by the regulation.

For example, the Provision and Use of Work Equipment Regulations 1998 state that an inspection record should be kept until the next inspection is made.

However, inspections of excavations, cofferdams and caissons, made in accordance with the Construction (Design and Management) Regulations 2015 (CDM), have to be kept on site until the works are completed, and after that for a further three months.

[illegible]

An example of a toolbox talk briefing record



An individual has three years to make a claim for neglect, act or omission where an injury or loss has been sustained.

11.4 Daily user and visual checks

Daily user and visual checks form a vital part of good health and safety site management, and reduce the risk of plant and equipment breakdown or failure, such as those listed below.

- Engine or motor seizure due to lack of lubrication.
- A low-pressure tyre being damaged, causing a vehicle to overturn.
- Missing scaffold edge protection increasing the likelihood of falling materials or persons.
- Damaged power plugs, sockets and leads, which could lead to electrocution.

Construction plant and equipment are exposed to harsh environments and they require effective maintenance regimes to prevent defects from developing. A programme of daily visual checks, regular inspections, servicing and maintenance schedules should be established according to the manufacturer's instructions and the risks associated with the use of each vehicle, piece of equipment or tool.

Plant hire companies must provide information with all plant and equipment that they supply to enable it to be used and maintained safely. Contractual arrangements between user and hirer should set out who is responsible for maintenance and inspection during the hire period and this should be made clear to all parties.

Plant and vehicles should have a maintenance log to help manage and record maintenance operations. Employers should establish procedures designed to encourage supervisors and operators to report defects or problems, and ensure that problems with plant and vehicles are put right. Planned inspection and maintenance needs to follow manufacturers' instructions.



Records of servicing, maintenance, inspections and examination should be kept up-to-date and available on site for all construction plant and equipment in use.

11.5 List of key requirements

The following is a list of key 'reporting and recording' requirements (for an explanation of these requirements refer to Appendix A).

Category	Brief description
1. Accident book BL510 (2018 edition)	Report of an accident must comply with the General Data Protection Regulation (GDPR).
2. F2508*	Report of an injury or dangerous occurrence.
3. F2508A*	Report of a case of disease.
4. F2508G1*	Report of a flammable gas incident.
5. F2508G2*	Report of a dangerous gas fitting.
6. F10	Notification of a project that falls under CDM.
7. F2067	Ionising radiation health record.
8. F2533	Safety representatives' report form.
9. F2534	Safety representatives' inspection form.
10. Air receivers	Safe working pressure and record of test.
11. Asbestos	Licence, notice and records. Where appropriate, maintaining a written asbestos management plan.
12. Builders' skips	On road, permits, lighting and signs.
13. Consultation with employees	Rights of employees.
14. Control of substances hazardous to health (COSHH) assessments	Maintenance of assessment forms and other records.
15. Danger areas	Identify with signs.
16. Dangerous substances and explosive atmospheres	Display of notices.
17. Diving	Records and registration.
18. Electrical equipment	Records, certification of new installations.
19. Electric shock placard	Display in workplaces.
20. Emergency evacuation	Emergency evacuation route signage as necessary and assembly point sign(s).

STATUTORY FORMS, NOTICES AND REGISTERS

Category (continued)	Brief description
21. Excavations, cofferdams and caissons	Records of inspections of places of work.
22. Explosives	Certificate and records.
23. Falsework	Records of design, including calculations, to be kept.
24. Fire	Fire risk assessment and safety plan. Notices for extinguishers, fire points and other fire-fighting equipment. Fire exit signs.
25. First aid	Notice regarding facilities and identity of first aiders.
26. Food hygiene	Display of notices, and certificates of staff training.
27. Fragile surfaces	Notice to be displayed (warning signs).
28. Hazardous substances	Labels on containers. Manufacturers' safety information, COSHH assessments.
29. Health and safety policy	Display or distribute to all employees.
30. Holes in floors and similar openings	Cover to be clearly marked. (Warning signs.)
31. Information for employees (see also Training)	Display approved Health and Safety Executive (HSE) law poster or distribute approved leaflet to all employees.
32. Insurance (employers' liability)	Display of current insurance certificate.
33. Ionising radiation	Advance notification to HSE of work. Licence, records and appropriate signage/ barriers at work site.
34. Lead	Records of inspection and thorough examination of control measures.
35. Lifting operations	Records of inspection, examination and safe working load (SWL) indication.
36. Management of health and safety	Risk assessments, appointments, procedures.
37. Manual handling	Assessment, marking of loads. Training.
38. Noise	Marking of hearing protection zones.
39. Plant and equipment	Inspection and records.
40. Pressure vessels	Display of working pressure, and other details.
41. Protective clothing and equipment	Assessments, maintenance, training.
42. Safety representatives and safety committees	Access to documents, minutes, and so on. Information on identity of trade union-appointed safety reps.
43. Safety signs	Signs and notices as appropriate.
44. Scaffolding	Records of inspections, display of 'incomplete' notice.
45. Training	Records.
46. Visual display units (VDUs) or display screens	Records of assessments, training, eyesight tests.
47. Waste management	Completion of duty of care documentation, forms, permits and certificates.
48. Work equipment	Suitability, maintenance, warnings.
49. Work in compressed air	Records of equipment inspection and health surveillance.
50. Working time	Permitted hours of work.
51. Working at height	Records of inspection of equipment.

**Only 'responsible persons' including employers, the self-employed and people in control of work premises should submit reports under the Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013. Responsible persons should complete the appropriate online report form.*

11.6 Statutory or recommended inspections and examinations

The following table sets out the recommended daily user checks, weekly inspections, statutory inspections and examinations.

Work activity plant item	Statutory or recommended						Form for statutory examination or report to comply with
	Pre-use daily	Weekly record	Monthly record	Three monthly record	Six monthly record	12 monthly record	
Excavations, cofferdams and caissons	✓ Inspect	✓ Inspect					Construction (Design and Management) Regulations 2015 (CDM)
Plant and equipment (not electrical or for lifting)	✓ Inspect	✓ Inspect			✓ Examine	✓ Examine	Provision and Use of Work Equipment Regulations 1998 (PUWER)
Plant and equipment (electrical) including fixed RCDs and portable 110 volt equipment	✓ Inspect	✓ Inspect		✓ Examine			Maintaining portable electrical equipment (HSG107)
Cranes and plant for lifting people, MEWPs, safety harness, lifting accessories and safety nets	✓ Inspect	✓ Inspect			✓ Examine	✓ Examine	PUWER, Lifting Operations and Lifting Equipment Regulations 1998 (LOLER)
Cranes and plant used for lifting	✓ Inspect	✓ Inspect				✓ Examine	LOLER
Work at height, all scaffolds, working platforms, mobile towers, ladders and steps, and similar items	✓ Inspect	✓ Inspect			✓ Examine		Work at Height Regulations 2005, PUWER, CDM
Fire-fighting appliances			✓ Inspect			✓ Examine	CDM, Regulatory Reform (Fire Safety) Order 2005
Site offices electrical equipment and installation		✓ Inspect				✓ Examine	CDM, Workplace Health, Safety and Welfare Regulations 1992

Note: this table is to be used only as a guide. A competent health and safety professional should confirm the frequency of the checks required.

STATUTORY FORMS, NOTICES AND REGISTERS – APPENDIX A

Appendix A – Brief explanation of key requirements

1. Accident book BL510	The keeping of an accident book is required by the Social Security (Claims and Payments) Regulations 1987, the Social Security Administration Act 1992 and the Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013 (RIDDOR). While there is no statutory book, the design of BL510 allows for an entry to be made without the personal details of a previous entry being seen. Any business that has its own accident book must ensure their version complies with the General Data Protection Regulation (GDPR). All accidents that cause any injury to an employee, no matter how slight, must be entered. Entry may be made either by the employee or anyone acting on their behalf. Completed book stubs and records must be kept for three years from the date of the last entry. An individual has three years to make a claim for neglect, act or omission where an injury or loss has been sustained.
2. F2508 Report of an injury or dangerous occurrence	Form F2508, report of an injury or dangerous occurrence, is required by RIDDOR. Notification should now be made online. The form will then be submitted directly to the RIDDOR database and a copy received back for record purposes. All incidents should be reported online but a telephone service remains for reporting fatal and specified injuries only.
3. F2508A Report of a case of disease	A report on form F2508A is required by RIDDOR. It must be completed when a registered medical practitioner has diagnosed in writing that an employee is suffering from a scheduled reportable disease and the person has been employed in a scheduled work activity by the employer.
4. F2508G1 Report of flammable gas incidents	A report is required online, via this form, if you are a distributor, filler, importer or supplier of flammable gas and you learn, either directly or indirectly, that someone has died or suffered a specified injury in connection with the gas that you distributed, filled, imported or supplied.
5. F2508G2 Report of a dangerous gas fitting	Gas installation businesses, registered with the Gas Safe register, must provide the HSE with details (using this form) of any gas appliances or fittings that they consider to be dangerous to such an extent that people could die or suffer specified injuries because the design, construction, installation, modification or servicing could result in: ■ an accidental leakage of gas ■ inadequate combustion of gas (such as increasing the proportion of carbon monoxide in appliance flue gases to a dangerously high level in normal or abnormal and adverse weather conditions) ■ inadequate removal of the gas combustion products.
6. F10 Notification of a project under the Construction (Design and Management) Regulations 2015 (CDM)	This form is required under the above regulations and must be filled out and submitted to the HSE online before the start of any construction work on a construction site. A project is notifiable if the construction work is likely to: ■ last for more than 30 working days and have more than 20 workers working simultaneously at any point in the project, or ■ exceed 500 person days.
7. F2067 Ionising radiation health record	The record, or a copy of it, must be preserved until the person to whom it relates has, or would have, attained the age of 75 years, but in any event for at least 50 years from the date of the last entry.
8. F2533 Safety representatives' report form	The Safety Representatives and Safety Committees Regulations 1977. This form is used by trade union-appointed safety representatives to notify the employer of any matter discovered during a workplace inspection, which is considered to be unsafe and/or unhealthy.
9. F2534 Safety representatives' inspection form	This form is used by trade union-appointed safety representatives to record the time, date and location of workplace health and safety inspections carried out.
10. Air receivers	The Pressure Systems Safety Regulations 2000. Every air receiver must have the safe working pressure clearly displayed on it. It must be examined and tested as laid down in the regulations and records kept.

11. Asbestos	The Control of Asbestos Regulations 2012. Virtually all work with asbestos requires a licence from the HSE or the giving of 14-days' notice of the intention to work with asbestos to the enforcing authority. The majority of work must only be undertaken by a licensed contractor. This work includes most asbestos removal, all work with sprayed asbestos coatings and asbestos lagging and most work with asbestos insulation and asbestos insulating board (AIB). Brief written records should be kept of non-licensed work, which has to be notified (for example, copy of the notification with a list of workers on the job), plus the level of likely exposure of those workers to asbestos. This does not require air monitoring on every job if an estimate of degree of exposure can be made, based on experience of similar past tasks or published guidance. Persons in charge of premises have a duty to manage the existence of asbestos in those premises, which will entail keeping records of surveys and possibly a register of where asbestos has been found, its condition and how its presence is being managed. This is known as an asbestos management plan. If an employee is exposed to asbestos above the control limit, health records of the employee must be maintained and kept for 40 years after the date of the last entry. All products containing asbestos must be properly labelled and recorded on the asbestos register. All waste must be in properly sealed and labelled containers.
12. Builders' skips	The Highways Act 1980 – Section 139. If a builder's skip is placed on a road, a permit must be obtained from the local highway authority and the requirements of the permit strictly followed. These will include ensuring the skip is properly lit during the hours of darkness. It must not pose a hazard to highway users and pedestrians but must be clearly and indelibly marked with the owner's name and telephone number or address.
13. Consultation with employees	The Health and Safety (Consultation with Employees) Regulations 1996 were introduced because the Health and Safety at Work etc. Act 1974 (HSWA) only covered union-appointed safety representatives. They now extend to all employees the right to have consultation with their employers and be provided with information. There are many similarities with the Safety Representative and Safety Committees Regulations, and they also implement the consultation provisions of the European Union Framework Directive on health and safety.
14. Control of Substances Hazardous to Health and Approved Code of Practice	The Control of Substances Hazardous to Health Regulations 2002 and the Approved Code of Practice (ACoP) require assessments to be made of substances hazardous to health and, except in the simplest and most obvious of cases, for the assessments to be written and kept accessible for those who need to know the results. If health surveillance is appropriate, the health records of employees under health surveillance must be maintained and kept for 40 years after the date of the last entry. All mechanical control measures (such as dust extraction) must be subject to routine examination in accordance with the regulations and records kept. Substances hazardous to health must be properly labelled.
15. Danger areas	Identify with signs under the relevant regulations.
16. Dangerous substances and explosive atmospheres	Under the Dangerous Substances and Explosive Atmospheres Regulations 2002, the following definitions and conditions apply. Dangerous substances A substance or preparation that is explosive, oxidising, extremely flammable, highly flammable or flammable, or any dust that can form an explosive mixture with air or an explosive atmosphere. Explosive atmosphere A mixture, under atmospheric conditions, of air and one or more dangerous substance in the form of gases, vapours, mists or dusts in which, after ignition has occurred, combustion spreads to the entire unburned mixture. Where an explosive atmosphere may occur, a specific sign is to be erected. In accordance with the Health and Safety (Safety Signs and Signals) Regulations 1996, the sign must consist of black letters on a triangular yellow background with black edging.
17. Diving	The Diving Operations at Work Regulations 1997. Records must be kept of the written appointment of all diving supervisors and of the qualifications and medical certificates of divers. All dives must be recorded in the diver's logbook. Diving rules have to be in writing. Diving is a specialised area, even in onshore waters. All diving contractors must be registered with the HSE.
18. Electrical equipment	The Electricity at Work Regulations 1989. All electrical equipment, including portable equipment, should be inspected on a regular basis by a competent person and records kept. Portable electric tools should be tested (portable appliance testing, or PAT) in accordance with HSE guidance.

STATUTORY FORMS, NOTICES AND REGISTERS – APPENDIX A

19.	Electric shock placard The Electricity at Work Regulations. Guidance notices or placards giving details of emergency resuscitation procedures in the event of an electric shock should be displayed in locations where people are at an enhanced risk of electric shock.
20.	Emergency evacuation CDM state there must be a sufficient number of emergency escape routes, which must lead as directly as possible to a point of safety. Escape routes must be indicated by signs and kept clear and free from obstructions.
21.	Excavations, cofferdams and caissons CDM require that excavations, cofferdams and caissons must be inspected and written reports of the inspections made.
22.	Explosives Virtually all possession of explosives requires an explosives certificate, which is issued by the local chief officer of police. Detailed records have to be kept of all movements or usages of explosives.
23.	Falsework and temporary works CDM and British Standard BS 5975 <i>Code of Practice for falsework</i>. These regulations include falsework under the definition of construction work and, therefore, the designer's duties apply. Temporary works require designing and can simply be defined as anything that is not part of the permanent works. It is recommended that records should be kept of all design calculations, drawings, estimated loadings and specifications for falsework, together with written permissions to pour concrete or to load falsework, and to dismantle it.
24.	Fire The Regulatory Reform (Fire Safety) Order 2005. Documentary information relating to fire safety should include: ■ a suitable and sufficient fire risk assessment to be carried out by a responsible person, or someone appointed to do so with sufficient experience and qualifications ■ records of staff training in the use of extinguishers ■ records of fire extinguisher servicing ■ records of practice evacuations ■ written fire risk assessment and a written fire safety plan ■ display of relevant signs where flammable and highly flammable substances are stored.
25.	First aid The Health and Safety (First-Aid) Regulations 1981 state: 'An employer shall inform his employees of the arrangements that have been made in connection with the provision of first aid, including the location of equipment, facilities and personnel.' The guidance notes recommend that first-aid notices are displayed as an effective means of informing the workforce of the employer's arrangement for first aid. There is no statutory form but a range of first-aid notices are available from the suppliers of workplace safety signs.
26.	Food hygiene The Food Safety Act 1990. Toilets adjacent to food rooms must be separated by a lobby. A notice stating 'Now wash your hands' must be displayed. Checks and inspections of equipment and staff training should be recorded. Certificates of staff training in food hygiene and handling must be displayed.
27.	Fragile surfaces A requirement of the Work at Height Regulations 2005 is that if people have to work near or pass close by any fragile roofing materials, the appropriate warning notices must be clearly displayed at all approaches to the area.
28.	Hazardous substances Generally, under the Control of Substances Hazardous to Health Regulations and the European Union Directive for Classification, Packaging and Labelling of Dangerous Goods, all containers containing hazardous substances should be clearly marked with their contents and the appropriate hazard warning symbol. Assessments must be made and, with minor exceptions, recorded (there is no statutory form). See also entries under Asbestos and Lead.

29.	Health and safety policy HSWA requires employers to have a safety policy and arrangements to bring it into force. If five or more persons are employed, that policy must be written down. There is no statutory form but the HSE publication <i>Writing your health and safety policy statement</i> may be used. The policy must be brought to the notice of all employees.
30.	Holes in floors and similar openings The Work at Height Regulations require that holes in floors and flat roofs, where a person is liable to fall through (referred to as <i>danger areas</i> in the regulations), must be properly protected. If a cover is used over a hole it must be clearly marked to show its purpose.
31.	Information for employees The Health and Safety (Information for Employees) Regulations 1989 require every employer to display the approved health and safety law poster in a position where it can be easily read by the employees, or to give to each employee a copy of the approved leaflet entitled <i>Health and safety law. What you should know</i>, which contains the same information.
32.	Insurance (employers' liability) The Employers' Liability (Compulsory Insurance) Act 1969 requires there to be in force insurance against liability for bodily injury or disease sustained by employees. The Employers' Liability (Compulsory Insurance) Regulations 1998 require a certificate of employers' liability insurance to be displayed at the place of business in such a position that it can be easily seen and read by employees, including every construction site. Employers can display an online version of the certificate, providing all employees have access to it. A hard copy may still be displayed.
33.	Ionising radiation Under the provisions of the Ionising Radiation Regulations 2017, any employer intending to use any source of ionising radiation must give 28-days' notice to the HSE. Very detailed records, including health records, have to be kept. Advice should be sought from the HSE or other competent source. The appropriate warning signs and notices for controlled areas must be displayed.
34.	Lead The Control of Lead at Work Regulations 2002, together with the Approved Code of Practice and guidance notes, require a number of forms to be completed by employers (and in some cases medical practitioners) if the required risk assessment under the regulations carried out by the employer shows that an employee's exposure to lead is liable to be significant. Some of the types of work likely to lead to significant exposure are: ■ high temperature work (over 500°C), including welding and cutting ■ abrasion of lead or lead-painted surfaces ■ spray painting with lead paint. Details of all the forms are given in the Approved Code of Practice.
35.	Lifting operations Under the Lifting Operations and Lifting Equipment Regulations 1998: ■ all machinery and accessories used for lifting are marked to indicate their safe working load (SWL) for each configuration in which they can be used ■ lifting equipment designed to lift persons is clearly marked as such ■ lifting equipment not designed for lifting persons but which could be easily mistaken for such is marked appropriately ■ all lifting equipment and accessories should be subjected to a periodic or scheme approach to thorough examination ■ records of thorough examination are made and kept available for inspection.
36.	Management of health and safety There is no statutory or recommended style of form for risk, or any other assessments. Your own devised form or a commercially obtained form may be completed either manually or on a computer database. Risk assessments must be made of all work activities. If you employ five or more people, the significant findings must be recorded and, in addition: ■ health surveillance – if needed, individual health records must be kept ■ emergency procedures – must be written down.
37.	Manual handling The Manual Handling Operations Regulations 1992. Assessment to be made where risks from manual handling cannot be avoided. It is recommended that all but the simplest assessments should be recorded.

STATUTORY FORMS, NOTICES AND REGISTERS – APPENDIX A

38. Noise	The Control of Noise at Work Regulations 2005 require that all hearing protection zones are identified by means of a sign (<i>refer to the Health and Safety (Safety Signs and Signals) Regulations</i>). A hearing protection zone is anywhere where an employee is likely to be exposed to a daily personal noise exposure of 85 dB(A) or a peak sound pressure of 137 dB(C).
39. Plant and equipment	If it is not otherwise provided for, it is strongly recommended that all plant, tools and equipment are subject to inspection and examination, and proper records kept. This will fulfil the employer's obligations under HSWA Section 2 and the Provision and Use of Work Equipment Regulations. A daily inspection and a six-monthly examination may be appropriate. Insurance companies may also require such routine inspections and examinations.
40. Pressure vessels	The Simple Pressure Vessels (Safety) Regulations apply to all pressure vessels intended to contain air or nitrogen at a greater pressure than 0.5 bar. Any such vessel first used after 1 July 1992 must have (amongst other things) details of the maximum working pressure, maximum and minimum working temperatures, and cubic capacity clearly displayed on it.
41. Protective clothing and equipment	The Personal Protective Equipment at Work (Amended) Regulations 2022. The following must be recorded: ■ risk assessment of the need for personal protective equipment (PPE) ■ PPE inspection, maintenance or results of any required tests ■ records of PPE issue and use.
42. Safety representatives and safety committees	The Safety Representatives and Safety Committees Regulations, and the Approved Code of Practice and guidance notes, give safety representatives the right to inspect and take copies of any document relevant to safety that an employer is required to keep (except health records of identifiable individuals). The employer must make available to them any health and safety information that would help safety representatives fulfil their functions. Where a safety committee has been established, proper minutes and records should be kept. Safety representatives may give written reports to management concerning safety in the workplace.
43. Safety signs	The Health and Safety (Safety Signs and Signals) Regulations. All signs giving health or safety information or instructions must comply with the relevant British Standard. A safety sign is anything that combines geometrical shape, colour and pictorial symbols to give safety information.
44. Scaffolding	■ Display of incomplete scaffold notice. Under the Work at Height Regulations, designated danger areas must be created where there is a risk of a person falling or being hit by a falling object. In the case of incomplete scaffolding, suitable notices must be displayed to discourage attempted access on to the scaffold. ■ Handover certificates. There are no statutory provisions requiring the use or production of scaffold handover certificates. However, all reputable scaffold contractors will issue them and they can be used to demonstrate that the scaffold was properly erected by competent persons and for the purpose for which the scaffold was designed. ■ Reports of inspections. To be completed and retained to comply with the Work at Height Regulations.
45. Training	Section 2 of HSWA, together with various other regulations made under the principal Act, require employers to provide information, instruction and training. It is most strongly recommended that all such information, instruction and training are properly and fully recorded so that employers are in a position to prove that duties under the Act or regulations have been met.
46. Visual display units (VDUs) or display screens	The Health and Safety (Display Screen Equipment) Regulations 1992. Suitable and sufficient analysis of workstations for the purpose of assessing health and safety risks. All but the simplest and obvious cases must be recorded. Entitles employees to eyesight tests. Records of requests and results are recommended.

47.	Waste management All contractors that carry or collect waste should have a waste carrier's licence and all waste disposal facilities should have a waste management licence or permit unless they are exempt. All waste transfers must be supported by the correct documentation, called a controlled waste transfer note (or, in the case of hazardous waste, a consignment note).
48.	Work equipment The Provision and Use of Work Equipment Regulations. Records must be kept, including risk assessment for selection and use, regular and statutory inspections, maintenance, training and instructions. All work equipment must be marked in a clearly visible manner where necessary, in the interests of health and safety. Warnings, audible or visible, to be incorporated into work equipment as necessary.
49.	Work in compressed air The Work in Compressed Air Regulations 1996. This is an extremely specialised area where expert knowledge is needed. Detailed records must be kept of, for example, the following areas. ■ Clinical records by a doctor. ■ Exposure records by a compressed air contractor. ■ Health records by an employer. These must be kept and maintained for 40 years from the date of the last entry.
50.	Working time The Working Time Regulations 1998 give details of permitted work hours and special consideration that should be given to young workers. Records of agreements, which can only last up to five years, must be kept by the employer.
51.	Working at height The Work at Height Regulations. This regulation specifies the need for competence in all persons involved in any capacity with regard to work at height. Any training carried out to achieve competency should be recorded. Where any person at work may pass across or near to a fragile surface, or actually work on it, prominent signs indicating that it is a fragile surface must be fixed at every approach to that place. Where any person could be injured by falling or being hit by a falling object, danger areas must be created to prevent such an occurrence. Danger areas must be clearly indicated, usually by signs and/or barriers. Where inspections of work equipment are carried out under this regulation, a record of the inspection must be made and retained as specified.



CONTENTS

Accident prevention and control

12

12.1	Introduction	146
12.2	Important points	147
12.3	Near-miss reporting	147
12.4	Accident and incident trends	148
12.5	The cost of accidents and incidents	149
12.6	HSE research into accident costs	149
12.7	Causes of accidents and incidents	149
12.8	Planning for health and safety	150
12.9	Factors to consider	150
12.10	Hazards	151
12.11	Implications of inexperience	152
12.12	Contractors and the self-employed	153
12.13	Supervision and control	153

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

ACCIDENT PREVENTION AND CONTROL

Overview

There are many factors that contribute to accidents in construction, their trends and causes, the impact accidents can have and the preventative measures that are required. This chapter is designed to help managers gain an appreciation of the factors that need to be considered when developing construction phase plans and safe systems of work, assessing risk, undertaking site inspections and managing health and safety on site each day.

12.1 Introduction

Accident statistics are a reliable indicator of the health and safety performance of the construction industry. The HSE publishes an annual statistical report of key statistics covering work-related ill health (all illness) and work-related injuries (fatalities and non-fatal injuries).

www This report highlights accident and ill-health trends, and is free to download from the HSE website.

The most recent construction industry statistics are given below.

- 45 fatal injuries to workers, and also five fatal injuries sustained by members of the public.
- An estimated 69,000 workers suffering from work-related ill health (new or long-standing cases): 54% were musculoskeletal disorders.
- 4,038 non-fatal injuries to employees were reported under the Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013. Of these, 1,539 (38%) were specified injuries, and 2,499 (62%) were over seven-day injuries.
- Falls from height are still the biggest cause of fatalities, with 51% of all fatal injuries being caused by falling from height.
- The total economic cost of injuries and new cases of ill health from current construction working conditions was estimated at £1.4 billion.
- Vibration white finger, carpal tunnel syndrome, occupational deafness and dermatitis are the most common cases of non-lung disease in the construction industry.

Around 2.6 million working days are lost each year.

- Working days lost through illness account for 2.1 million.
- Working days lost through injury account for 0.5 million.

www View the latest health and safety statistics detailing the kinds of injuries that have occurred in construction on the HSE website.

Each year the plan of work for the construction division of the HSE acknowledges that some parts of the industry have made significant steps forward over recent years; this is particularly noticeable on larger projects. However, these improvements are not mirrored to anything like the same extent on smaller sites, where many instances of unacceptable standards can still be found.

www For details of the HSE's plan of work visit the HSE website.

This diagram is a version of **Bird's triangle**. Its aim is to demonstrate the approximate relationships between the different levels of accident that occur. It shows that for each fatality there are on average 10 serious injuries, 30 accidents with injury, 600 incidents without injury (near misses or property, equipment or material damage) and thousands of unsafe acts or conditions.

Bird's research illustrates that accident prevention should not be about just concentrating on fatalities and serious injury, but on preventing unsafe acts and conditions. Near misses can be understood as an early indication that unsafe acts or conditions are happening, rather than reacting to actual accident and injury trends. This is why near-miss reporting and proactive hazard-spotting techniques (such as pre-use daily checks, workplace inspections and audits) are so important.



An unsafe act or near miss under a slightly different set of circumstances could be a serious or even fatal accident. This makes prevention more difficult, particularly if incidents go unreported. Many companies have found that their accident rate has reduced as the number of reported incidents with no injury or near misses has increased, through management encouragement for such incidents to be reported.

12.2 Important points

- The construction industry remains a high-risk industry and consistently accounts for a high number of fatalities and accidents.
- The construction industry accounts for about 6% of employees in Great Britain.
- Accident prevention has to be actively managed; a good safety record will not just happen and everyone has their part to play.
- Reported details of accidents show that in the vast majority of cases the accident could easily have been prevented by taking simple precautions.
- The HSE has stated that many accidents result from decisions taken before work actually starts (planning and identifying safe systems of work).
- You may have no influence over these decisions but find that you need to challenge the health and safety implications that arise as a consequence of them.
- The true cost of an accident goes beyond the financial implications.
- Statistics show that new starters on site, and those at both ends of the age spectrum, are the most prone to accidents.



Accident prevention

- An **accident** is an unplanned, unscheduled, unwanted event or occurrence, or any undesired circumstance, that may result in injury to persons and damage to property. The injured person may not be an employee and damaged property may not belong to an employer.
- **Hazard** is the potential to cause harm, including ill health and injury; damage to property, plant products or the environment; production losses or liabilities.
- **Risk** is the likelihood that a specified, undesired event will occur due to the realisation of a hazard by or during work or by products created by work. The severity of harm is also considered when calculating the risk level.
- **Incident** is another word that is sometimes used to describe an accident. The main difference is that an incident is something that happened that may or may not have resulted in an injury or damage.

12.3 Near-miss reporting

The importance of learning from experience cannot be overstated. It is an essential element of accident prevention.

A near miss is an incident that had the potential to result in personal injury and/or damage to the structure under construction, plant and equipment or the environment.

Individual companies will decide on their criteria for categorising an incident as a near miss.

Unsafe conditions - something with the potential to cause harm.

Near misses - an incident that nearly resulted in an injury, damage or loss.

Accidents - an accident that resulted in an injury, damage or loss.

The details of all near misses must be accurately and honestly reported to enable the circumstances to be investigated and measures put in place to prevent a recurrence.

To achieve an effective near-miss reporting system, the following are required of the workforce.

- To trust that management will treat the incident fairly and objectively.
- Be encouraged to report near misses with the assurance that individuals involved will not be disadvantaged by their honesty.
- Have confidence that the issues raised will be addressed, or else 'why bother?'.
- Be provided with the means of promptly recording the details of exactly what happened and offering their opinion as to why it occurred.

Companies may find it is beneficial to provide easy to access near-miss reporting forms or cards, which can be completed in privacy and anonymously if that is the individual's choice.

However, anonymous reporting does not provide the opportunity for follow-up discussions to establish more details, and it may encourage malicious reports to be submitted.

Evidence shows that near-miss reporting linked to recognition or a reward scheme has the best chance of succeeding.

ACCIDENT PREVENTION AND CONTROL

12.4 Accident and incident trends

Broad categories are used to classify all accidents that are reported to the HSE.



Reportable accident statistics for the construction industry show that the same types of accident happen frequently, year after year. Those that continue to figure prominently in the statistics include: falls from a height; being trapped by something collapsing or overturning; struck by moving (including flying or falling) objects; slips, trips or falls on the same level; and being injured while handling, lifting or carrying.

Further challenges may have been added as the competition for contracts has increased and tendered margins decreased. Commercial pressure may tempt contractors to take increased risks with their employees' health and/or safety.

The stubborn resistance of the accident statistics to significantly reduce is often attributed to the difficult working conditions experienced on construction sites, where the environment of the workplace is continually changing as work progresses.

However, in reality, it is often a lack of training or health and safety awareness among construction workers, or a lack of effective health and safety management, that prevents a reduction in accidents.

12.5 The cost of accidents and incidents

12.5.1 Cost to the victim and their family

- Pain and suffering.
- Loss of earnings.
- Extra expense.
- Continuing disability and depression.
- Incapacity for the same job.
- Incapacity for work outside the job.
- Effect on dependants and friends.

12.5.2 Cost to people directly responsible

- Worry and stress.
- Recriminations and guilt.
- Extra work (reports, training and recruitment).
- Loss of credibility.

12.5.3 Cost to the company

- Working time lost by victim.
- Time lost by other employees out of sympathy, curiosity or discussion.
- Time lost by supervisors and others investigating the accident.
- Possible damage to machines or materials.
- Idle time (replan, repair and reinstate job).
- Rise in insurance premiums and cost of prosecution or civil action.
- Damage to reputation and possible failure to obtain work.

12.5.4 Cost to the working group

- Shock, disbelief, anger and personal grief.
- Low morale and negative effects on production.

12.5.5 Cost to the nation

In social and economic terms, accidents are an unwanted expense.

- Hundreds of thousands of person-days are lost each year.
- Industrial accidents have an impact on NHS resources.
- Millions of pounds are paid in pensions and death benefits.
- Countless lives are changed for the worse.

It must be emphasised that these are all negative impacts.

Good health and safety management prevents accidents, which means fewer or no victims or affected families, no additional costs to companies, no loss of reputation and no loss of work.

12.6 HSE research into accident costs



HSE Research Report No. 464 contains a number of accident case studies that highlight the cost of accidents to the victims, in the widest terms.

Several of the case studies feature accidents to construction workers.

The consequences of each accident are considered from different standpoints.

Vocational – impact on future job prospects.

Economic – significant loss of income.

Social – standard-of-life issue for victim and family.

Behavioural – reliance on medication, inability to concentrate, inability to sleep, ill-temper, and so on.

Psychological – mood swings, loss of memory, emotional instability and guilt.

Examples of the impact that the accidents had on the victims and on family members, as extracted from the construction case studies, are listed below.

- 'I had no realistic prospect of returning to work in the immediate future'.
- 'I received no support from company or colleagues upon returning to work other than "light duties" for the first two weeks'.
- 'My family was reliant on state benefits of £66 per week during three and a half months off work'.
- 'I am unable to bathe or dress myself, have problems with walking and no longer visit the gym'.
- 'I am unable to sleep due to discomfort, increased eating and inability to concentrate'.
- 'Years later I still suffer flashbacks and nightmares'.

12.7 Causes of accidents and incidents

Examining accident details will help to establish common factors and trends, revealing any weaknesses in a company's health and safety management system.

Accidents can be caused by unsafe acts and attitudes or behaviours of people at work, which result in unsafe conditions being created.

They are also caused by a lack of foresight or planning, which may be a failure to set up a safe system of work, or failure to appreciate the results of risk assessments, control of substances hazardous to health assessments or other similar work.



Unsafe acts can create unsafe conditions, which can cause accidents. These accidents often result in injury or damage.

It is impossible to list all the different types of unsafe acts and unsafe conditions that can exist in the construction industry.

However, it is worth recording the most frequent known causes of accidents on construction sites, as follows.

- Lack of planning, management control and supervision.
- Lack of knowledge of good safety techniques or lack of safety awareness.
- Unsafe methods of working at height, including lack of provision for fall prevention.
- Incorrect or unauthorised use of machinery and equipment.
- Failure to segregate operating plant and pedestrians.
- Failure to inspect and maintain all types of machinery and equipment.
- Incorrect use of tools and equipment (hand tools and power tools).
- Use of faulty equipment with improvised repairs or modifications.
- Unsafe manual handling (lifting, loading, moving, stacking and storing).
- Overloading of working places (scaffolds, falsework, hoists, machines, vehicles and roofs).
- Failure to use protective safety equipment (safety helmets, eye protection, gloves, respiratory protective equipment and footwear).
- Carrying out work on moving parts with guards removed or safety devices inoperative.
- Failure to report faulty or unsafe equipment, or dangerous occurrences and incidents.
- Creating unstable structures.

ACCIDENT PREVENTION AND CONTROL

12.8 Planning for health and safety

Despite the effort made by the majority to fulfil their legal, moral and social obligations, difficulties are often encountered in human behaviour that require time and tolerance before acceptable health and safety standards are achieved. It is essential that careful consideration is given to pre-planning, communication, training, supervision and consultation with the workforce, if safe systems and safe and healthy places of work are to be developed and maintained. All of the following measures can make a significant contribution towards the prevention of accidents through the implementation of safe systems of work and procedures.

- Allowing enough resources (time and financial).
- Establishing procedures for communication and consultation with the workforce.
- Providing adequate protection and guarding of working places, platforms, machinery, tools, plant and equipment.
- Implementing an adequate system for the maintenance and repair of plant, equipment and tools.
- Bringing about and maintaining an awareness of, and compliance with, all safety legislation and information relating to systems and procedures of work.
- Providing appropriate training, instruction and information at all levels, including safety training.
- Providing adequate supervision and control.
- Planning, siting and/or stacking materials and equipment to allow safe access or egress of site plant, vehicles and equipment.
- Pre-planning and organising site layout, to provide maximum efficiency, safety and progression of the work sequences and operations.
- Providing adequate resources and equipment to protect and maintain the health and welfare of all personnel.
- Producing, declaring, maintaining and supporting a health and safety policy, and updating it as appropriate to accommodate advancement and development.



The safe segregation of construction plant and pedestrians avoids accidents



Organised site layout, to provide maximum safety

12.9 Factors to consider

Health and safety at work will be affected by human and personal factors, job factors and environmental factors.

12.9.1 Human and personal factors

Attitudes of people at work often play an important part in the prevention of accidents, and conversely, a wrong attitude can cause accidents to happen. Attitudes differ depending on the characteristics of the person.

- Age and gender.
- General health, natural dexterity, agility, physique and ability.
- Disabilities or medical conditions, if any.
- Senses of smell, sight, hearing, touch and, sometimes, taste.
- Education and qualifications, training and skills.
- Home and social life, status at home and work and position in peer group.



Safe attitudes = safe actions = safe conditions.



For further information refer to the HSE publication *Reducing error and influencing behaviour* (HSG48).

12.9.2 Job factors

Every work activity has a degree of inherent hazard. Building and construction sites can be particularly hazardous and demand the co-ordination of a large number of trades, skills and work at any one time.

Particular attention should be given to the following.

- The adequacy of time and resources to plan the job and to do the job.
- Provision of tools and equipment that are safe to use, inspected regularly and properly maintained.
- Implementation of safe systems of work.
- Personnel who are unfamiliar with established safe systems of work and practices.
- Personnel who are new to a specific worksite or unfamiliar with a new working environment.
- Those lacking induction training and/or experience.
- The provision of adequate training, information and supervision.
- Balanced workload, fatigue and boredom.

12.9.3 Environmental factors

The majority of people do not work in isolation.

The attitudes of others in a working group (for example, managers, supervisors, safety advisers, safety representatives and shop stewards) may help to prevent accidents, but the following should also be considered.

- The accident record of the firm, site and working group.
- The inter-relationship of people within the group.
- Information and communication processing methods.
- Weather conditions (hot, cold, wet, wind, ice or snow).
- Working at heights, in confined spaces or underground.
- Working conditions (noise, dust, light and ventilation).
- Health and welfare facilities.

12.10 Hazards

12.10.1 Obvious dangers

- Vehicles not being segregated from pedestrians.
- The presence of highly flammable material and other fire hazards.
- Work platforms or work places at height with no fall prevention.
- Insecurely stacked, slung, lifted and transported loads.
- Unsafe machinery, equipment and tools.
- Unsafe working areas due to weather conditions.
- Confined spaces or unsupported excavations.

12.10.2 Potentially dangerous situations

Below are some examples of circumstances that might result in an accident (this list is not fully exhaustive).

- Working with machinery or tools with guards or fences removed.
- Using the incorrect type of plant, tools or equipment for the work involved.
- Unauthorised removal of guard-rails, or failure to replace them following removal for access of plant or materials.
- Transport of insecure or unstable loads.
- Dropping tools and materials from a height.
- Failure to wear personal protective equipment or respiratory protective equipment.
- Working in unstable excavations, without adequate supervision and control.
- Untidy working places or congested walkways and areas, creating a tripping/safe exit hazard.
- Working at height or over water without edge and/or personal protection.
- Inadequate, incorrect or badly placed lighting.
- Unsafe electrical equipment, underground services and overhead cables.

ACCIDENT PREVENTION AND CONTROL

12.10.3 Operational risks

Examples of work requiring competence, careful monitoring and/or close supervision are listed below.

High risk activities	Operations creating health hazards or risk of injury
■ Demolition ■ Working at height ■ Excavations ■ Lifting operations ■ Work in confined spaces ■ Timber, concrete or steel frame erection ■ Roof work and cladding ■ Work associated with live traffic ■ Work around power lines/electricity	■ Work producing dust or fumes ■ Jobs with continual high exposure to noise or vibration ■ Jobs with continuous elements of the same type of manual handling (such as block laying and kerb laying) ■ Work with asbestos ■ Work with hazardous substances (respiratory, eye splash and skin)

12.10.4 Site security

Owners or occupiers of sites have a common law duty under the Occupiers' Liability Act 1957 to protect the health and safety of **anyone** who might enter the site, even if that entry is unlawful. This duty is also stipulated in the Construction (Design and Management) Regulations 2015 and the contractor or, where applicable, the principal contractor must take the necessary steps to prevent unauthorised access to site.

Contractors must not commence work until reasonable steps have been taken to prevent unauthorised access. This includes unauthorised access by members of the public, particularly young children (who in some cases have been killed or injured), who may trespass on the site, either during or outside of normal working hours.

Accidents have been caused by falls into holes, being trapped by earth, drowning, accidents involving vehicles and those involving falls and falling materials. In all these cases, better site security and management might have prevented the accident. Contractors must take all reasonable and practical steps to ensure that sites are secure.



For further information refer to the HSE publication *Protecting the public: Your next move* (HSG151).

The guidance focuses on the reality that hazardous construction work can be a fatal danger, not only for those within the industry but for the wider public too.

12.11 Implications of inexperience

12.11.1 Young persons



A young person is anyone under the age of 18.

Before employing a young person, the employer must assess the risks to the young person's health and safety, arising from the work they are required to do, in accordance with the relevant regulations. This assessment must take account of a number of factors.

- The inexperience and immaturity of young persons, and their lack of awareness of risks.
- The type of any work equipment involved and the way it is used.
- The potential for exposure to physical, biological and chemical agents.
- Any health and safety training that is required for young persons.

Having carried out this assessment, employers must then determine whether the level of risk has been reduced to as low as is reasonably practicable. There is particular importance placed on avoiding work that:

- is beyond the young person's physical or psychological capacity
- involves harmful exposure to agents that are toxic or carcinogenic, causes heritable genetic damage or harm to an unborn child or which in any way chronically affects human health
- involves harmful exposure to radiation
- involves the risk of accident, which it may be reasonably assumed cannot be recognised by young people owing to their insufficient attention to safety or lack of experience or training
- involves exposure to physical agents (such as extreme cold or heat, noise and vibration).

Consideration to the level of acceptable risk may be given for young persons over the minimum school-leaving age, where the work is necessary for their training, and where they are properly supervised.

12.11.2 New starters

New starters in a company or on a site, and inexperienced persons of any age, have similar problems to those of young workers.

They are subjected to a new environment, rules, methods and procedures; they are also under different supervision, are working with new colleagues and are using a variety of tools, equipment and manual effort to produce the work required. The start of their health and safety training is usually an induction into the company. The need for refresher and continuance training should be reviewed at intervals and carried out as necessary.

12.11.3 Older workers

There is legislation in place that makes it illegal to discriminate against older workers. It is also interesting, however, to note that the number of injuries to older workers is higher than average.

There are various reasons that have been suggested as to why.

- Overfamiliarity with the job.
- General slowing of reactions.
- General loss of strength and flexibility.
- Pre-existing damage to body and systems.
- Age-related degeneration of hearing and eyesight.

What is also noteworthy is that when an older person is injured, often the recovery time is longer, because the injury is more severe than it would be for a younger person.

12.12 Contractors and the self-employed

Problems may arise if accident prevention and reporting measures are not communicated to contractors and self-employed persons on site. Arrangements must be made to ensure that contractors and the self-employed are acquainted with, and adhere to, the principal contractor's health and safety standards and procedures, or any other policy or special instructions that may be in force, or relevant to specific operations.

The following recommendations are not exhaustive. Pre-planning at the tender stage may reveal additional requirements.

- Discussion, as necessary, should take place to ensure complete understanding or clarification of what is required.
- A listing of site personnel (who's who and details of established channels of communication).
- The availability, and validity, of certificates and documentation relating to health and safety knowledge and trade competency, including Construction Skills Card Scheme (CSCS), Construction Plant Competence Scheme (CPCS) and Construction Industry Scaffolders Record Scheme (CISRS) cards.
- Safe systems or methods and sequences of work should be established, particularly where such work may affect the health and welfare of others.
- Attention should be given to the requirements for the delivery of plant or equipment, transportation, loading and unloading of materials and provision of means of access, particularly when this work takes place out of normal working hours.

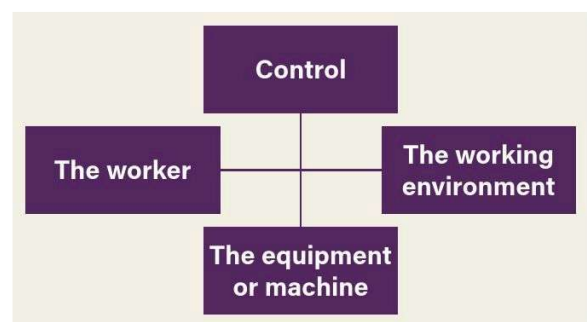
12.13 Supervision and control

The accident trend can be strongly influenced by providing adequate training and supervision to control the worker, the machine or the equipment and the working environment.

12.13.1 The worker

The following points are essential requirements for the worker.

- Be adequately trained and informed of the work they are expected to do.
- Be aware of all the hazards in any activity they are expected to do.
- Be competent to do the work or be under adequate suitably qualified supervision.
- Adopt a safe system of work and use the protection provided.
- Be aware of accident and emergency procedures.
- Be aware of the company's health and safety policy, rules and legislation **applicable to the work**.



Accident prevention is the control of these factors

ACCIDENT PREVENTION AND CONTROL

12.13.2 The working environment

This applies to all areas of the site, including the work area, stores, offices, depot and welfare facilities.

Regular checks are essential to ensure that these areas remain safe and are not a risk to health.

12.13.3 The equipment or machine

Ensure that the following procedures and practices are observed.

- Regular inspections are completed by trained and competent persons.
- No defective equipment is used.
- Defects are properly rectified.
- Adequate servicing and maintenance is carried out.
- Records and reports are maintained.
- All moving parts are adequately guarded or protected.
- Manufacturers' literature and instructions are available for operatives.
- Proper handling, lifting and slinging of equipment is conducted.
- Equipment and machines are adequately secured, both when in use and parked.
- Hand tools are inspected and maintained.



For further guidance refer to Chapter A10 Inspections and audits.

CONTENTS

Accident reporting and investigation

13

13.1	Introduction	156
13.2	Important points	156
13.3	Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013	156
13.4	Accident records	160
13.5	Calculating the incidence and frequency rates of accidents	160
13.6	Post-accident investigation	161
	Appendix A – Accident/incident reporting matrix	165

Supporting INFORMATION

GT700 Toolbox talks/supporting advice and guidance

Toolbox talks on some of these topics are available in the GT700 publication. Supporting advice and guidance covering some of these topics is available on our companion website.



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

ACCIDENT REPORTING AND INVESTIGATION

Overview

The recording of incidents and accidents at work is important, to help establish what went wrong and to prevent similar incidents and accidents from recurring.

This chapter outlines why and how details of incidents and accidents should be captured, what has to be reported to the Health and Safety Executive (HSE) and how to conduct an investigation.

13.1 Introduction

It may be said that there is no such thing as an accident and without people there are no accidents. An accident can always be put down to human error but it is by no means always the fault of the injured person. In fact, the HSE considers that a lack of safety management, or procedures, is often a contributory factor in a high percentage of accidents, particularly within construction.

The reporting of certain types of accident is a legal requirement and failure to comply is a criminal offence. The reports that are made allow the HSE to identify accident trends, and to take remedial action by producing additional health and safety guidance and legislation, where it is deemed necessary.

If you are an employer, self-employed or are in charge of work operations, you have duties under the law. You have to report deaths, specified injuries, non-fatal accidents requiring hospital treatment to non-workers and dangerous occurrences immediately – the HSE should be in receipt of a report within 10 days of the incident. Accidents resulting in injuries lasting for more than seven days have to be notified to the HSE within 15 days of the incident. Occupational diseases should be reported as soon as the employer receives the diagnosis.

13.2 Important points

- It is important that all workplace accidents, no matter how minor, are reported to the injured person's employer, site manager or supervisor as appropriate.
- There is a legal requirement for details of all accidents to be recorded.
- Once completed, the record of an accident is subject to the General Data Protection Regulation (GDPR) and is confidential.
- Certain types of accident, cases of occupational disease and some dangerous occurrences must be reported to the HSE.
- Each company or organisation should have a procedure for investigating workplace accidents.
- The investigation of accidents will enable trends to be established and preventative measures put in place.
- The level of investigation should be proportionate to the seriousness of the accident.

13.3 Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013

The Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013 (RIDDOR) require the following to be reported directly to the appropriate enforcing authority (either the nearest HSE office or the Local Authority).

- Deaths and specified injuries.
- Over seven-day injuries.
- Specified occupational diseases.
- Dangerous occurrences.

The regulations place a duty on the responsible person to make reports to the enforcing authorities. In the case of injuries to employees, the responsible person will be the employer. In the case of the victim being self-employed and working in domestic premises, or being informed by a doctor that they have a work-related disease or conditions, they would be the responsible person and should make the report themselves. If they are working in someone else's work premises, then the person in control of the premises is responsible for reporting. If a member of the public is the victim, then the responsible person will be the people in control of the work premises where the event occurred.

All sub-contractors must notify both the enforcing authority and the main/principal contractor of any reportable accidents. Where an accident, occupational disease or dangerous occurrence takes place that requires reporting under RIDDOR, a report should be made online via the HSE website. The RIDDOR section of the HSE website contains relevant guidance for completing the forms, to make submitting them easier.

- 2508 Report of an injury.
- 2508G1 Report of a flammable gas incident.
- 2508 Report of a dangerous occurrence.
- 2508G2 Report of a dangerous gas fitting.
- 2508A Report of a case of disease.

The following must be reported immediately to the appropriate authority by the quickest practicable method. All incidents must be reported online but a telephone service remains for reporting fatal and specified injuries only. The completed report must be submitted on the approved form within **10 days**.

- Death of any person as a result of an accident at work.
- An accident to any person at work resulting in specified injuries or serious conditions specified in the regulations.
- Any one of the dangerous occurrences listed in Schedule 2 of the regulations.
- Non-fatal accidents requiring hospital treatment to non-workers.

Over seven-day injuries must be reported on an approved form within **15 days**.

13.3.1 Why report and record?

The report informs the enforcing authorities (the HSE, Local Authorities and Office of Rail and Road (ORR), where relevant) about deaths, injuries, occupational diseases and dangerous occurrences, so they can identify where and how risks arise and whether they need to be investigated. This allows the HSE, Local Authorities and ORR to target their work and provide advice about how to avoid work-related deaths, injuries, ill health and accidental loss.

Records of incidents covered by RIDDOR are important. They ensure that you collect the minimum amount of information to allow you to check that you are doing enough to ensure safety and prevent occupational diseases. This information is a valuable management tool that can be used as an aid to risk assessment, helping to develop solutions to potential risks. In this way, records also help to prevent injuries and ill health, and control costs from accidental loss.

13.3.2 What must be reported?

13.3.2.1 Work-related accidents

Reportable injuries (including deaths) must be reported if they occur as the result of a work-related accident and the type of injury is reportable, as listed under types of reportable injuries. For the purposes of RIDDOR an accident is a separate, identifiable, unintended incident that causes physical injury. This specifically includes acts of non-consensual violence to people at work.

The HSE defines a work-related accident as an accident 'arising out of/in connection with work', and this means that an accident may still need to be reported even if there was no breach of health and safety law, and no-one was clearly to blame. When deciding if a report needs to be made, the key issues to consider are:

- What work was going on at the time?
- What was the injured person doing?
- What were others doing?
- Where did the accident happen?
- Were factors such as structures, equipment or substances involved?

An accident taking place at a work premises does not automatically mean that it was work related. The work activity itself must cause the accident, and that would be the case if any of the following were involved in the accident.

- How the work was carried out, including how the work was organised, supervised or performed by an employer or their employees, or by a self-employed person.
- Any machinery, plant, substances or equipment used in connection with the workplace or work processes.
- The condition of the workplace, including the state of the structure or fabric of a building or outside area forming part of the workplace, and the state and design of floors, paving, stairs or lighting.
- Where did the accident happen?
- Were factors such as structures, equipment or substances involved?



Guidance and examples of incidents that do and do not have to be reported are available on the RIDDOR section of the HSE website.

13.3.2.2 Deaths

A death to workers and non-workers must be reported if it results from a work accident or an act of physical violence to the worker.



Where an employee, as a result of an accident at work, has suffered a reportable injury that has caused their death within one year of the date of that accident, the employer shall inform the relevant enforcing authority, in an appropriate manner, of the death as soon as it comes to their knowledge.

ACCIDENT REPORTING AND INVESTIGATION

13.3.2.3 Injuries to people at work

RIDDOR gives two types of injury that must be reported if the person was at work – specified injuries and over seven-day injuries.

Specified injuries include the following.

- A fracture, other than to fingers, thumbs and toes.
- Amputation of an arm, hand, finger, thumb, leg, foot or toe.
- Permanent loss of sight or a reduction of sight.
- Any injury arising from working in an enclosed space leading to hypothermia, heat-induced illness, resuscitation or admittance to hospital for more than 24 hours.
- Unconsciousness caused by asphyxia or head injury.
- Crush injuries leading to internal organ damage.
- Serious burns (covering more than 10% of the body or damaging the eyes, respiratory system or other vital organs).
- Scalping (a separation of the skin from the head) requiring hospital treatment.

Over seven-day injuries are where an employee, or self-employed person, is away from work or unable to perform their normal work duties for more than seven consecutive days (not counting the day of the accident, but including weekends and rest days).

Where the worker's injury or condition does not become apparent until some time after the accident, it must be reported as soon as it has prevented them from doing their normal work duties for more than seven consecutive days. If the injured person would not normally have been expected to work during those seven days, you must take those days into account when deciding whether they were unable to do their normal duties for more than seven consecutive days.

13.3.2.4 Injuries to non-workers

You must report injuries to members of the public or people who are not at work, if they are injured through a work-related accident and are taken from the scene of the accident to hospital for treatment to that injury. Examinations and diagnostic tests do not constitute treatment in such circumstances and there is no need to report incidents where people are taken to hospital purely as a precaution when no injury is apparent.

13.3.2.5 Reportable occupational diseases

Employers and self-employed people must report cases of certain diagnosed reportable diseases (see below), and these must be linked to exposure to specified hazards at work. They should only be reported once you receive a confirmed diagnosis of the worker's condition. All of the diseases and hazards have to meet certain criteria to be reportable, and this is laid out in detail in the RIDDOR section of the HSE website. As an example, cramp of the hand or forearm would not be reportable if it's a one-off incident during someone's work activity.

- Hand-arm vibration syndrome (HAVS) (due to operating hand-held vibrating tools and equipment).
- Occupational dermatitis, due to work substances (such as cement, solvents and epoxy resins).
- Carpal tunnel syndrome.
- Cramp of the hand or forearm.
- Tendonitis or tenosynovitis of the hand or forearm.
- Any occupational cancer (such as mesothelioma, asbestosis (due to work with or exposure to asbestos) and lung cancer).
- Occupational asthma, where the person's work involves significant or regular exposure to a known respiratory sensitiser.
- Any disease attributed to an occupational exposure to a biological agent.



Occupational deafness (noise induced hearing loss) is not reportable under the list of occupational diseases, as it is hard to prove the cause of acquired hearing loss (deafness); for example, it could have been caused by any past work activity with exposure to high levels of noise, or through noisy hobbies undertaken outside of the working day.

13.3.2.6 Reportable dangerous occurrences

Dangerous occurrences are certain, listed, near-miss events. Not every near-miss must be reported. There are 27 categories of dangerous occurrences that are relevant to all workplaces. Some examples are shown below.

- The collapse, overturning or failure of load-bearing parts of lifts and lifting equipment.
- Plant or equipment coming into contact with overhead power lines.
- The accidental release of any substance that could cause personal injury, other than through combustion and flammable liquid or gases.



A full list of dangerous occurrences applicable to all workplaces, and additional categories of dangerous occurrences, can be found on the RIDDOR section of the HSE website.

13.3.2.7 Reportable gas incidents

If you are a distributor, filler, importer or supplier of flammable gas and you learn, either directly or indirectly, that someone has died, lost consciousness or been taken to hospital for treatment to an injury arising in connection with the gas you distributed, filled, imported or supplied, this should be reported online using F2508G1.

In practice, the emergency service providers representing gas conveyors usually carry out reporting duties.

If you are a gas engineer, registered with the Gas Safe register, you must provide details of any gas appliances or fittings that you consider to be dangerous to the extent that people could die or suffer a specified injury (this should be done online using form F2508G2).

These dangers may be due to the design, construction, installation, modification or servicing, and could result in the following.

- An accidental leakage of gas.
- Inadequate combustion of gas.
- Inadequate removal of products of the combustion of gas.



For an accident and incident reporting matrix refer to Appendix A.



Some examples to provide clarification

- A directly employed person breaks their arm at work. This must be reported by the employer, in their capacity as a responsible person, as a specified injury.
- A self-employed sub-contractor breaks their leg at work. The injury must be reported as a specified injury by the main/principal contractor acting in their capacity as the responsible person who was in control of the premises.
- An employee of a sub-contractor is informed by their doctor that they are suffering from work-related vibration white finger and subsequently informs their employer. The employer, in their capacity as a responsible person, must report the incident to the HSE as a reportable disease.
- An employee inadvertently hits an underground electric cable whilst operating a road-breaker. There is minor damage to the external sheath, but the conductor is not exposed. This is not reportable as there was not an electrical short circuit with fire and explosion. The incident, however, warrants significant internal investigation.
- A member of the public is knocked down by a lorry entering the site as it crosses the pavement. They are taken to hospital by ambulance and it is obvious they have suffered an injury. This would be reportable as it involves a member of the public being taken directly to hospital for treatment.
- A sub-contracted employee burns their hand and is taken to the accident and emergency department of the local hospital. The employee is back on site later that afternoon and continues to work as normal for the rest of the week. This would not be reportable. However, had the employee suffered more than 10% burns, the incident would be reportable.
- An employed delivery driver twists their ankle when they step down from their cab. They receive first aid, insist they are fit to drive and later leave the site. They subsequently take the following seven days off because of pain and swelling in their ankle. The incident should have been recorded in the site accident book, but it would seem unreasonable for the site to be aware of the consequence. The delivery driver's employer would have a responsibility to report this as an over seven-day injury.
- An employee sustains a head injury as a result of tripping over debris left on site. The accident occurred on a Friday and, because of the injury, the person is unable to return to work until a week later, on the following Monday. Although only five actual working days have been lost, the accident must be reported as an over seven-day injury because the Saturday and Sunday also count, as the injured person would have been unfit for work had these been working days.

13.3.3 How to report



Complete the appropriate online form, available from the HSE website. All incidents must be reported online, but a phone service remains through the Incident Contact Centre for reporting fatal injuries, specified injuries and major incidents only (+44 (0) 151 922 9235).

13.3.4 Reporting out of hours

The type of circumstances where HSE may need to respond out of hours include:

- following a work-related death
- following a serious incident where there have been multiple casualties
- following an incident which has caused major disruption, such as evacuation of people, closure of roads, or large numbers of people going to hospital.

ACCIDENT REPORTING AND INVESTIGATION

13.4 Accident records

Employers are legally required to keep records of all accidents. These can be stored in any medium, including electronically, as long as printable copies are readily available if required. You must keep a record of the following.

- **Accident, occupational disease or dangerous occurrence** that requires reporting under RIDDOR.
- Other occupational accident causing injuries that result in a worker being away from work or **incapacitated for more than three consecutive days** (not counting the day of the accident but including any weekends or other rest days). If you are an employer, who must keep an accident book under the Social Security (Claims and Payments) Regulations 1979, that record will be enough.

You do **not** have to report over three-day injuries, unless the incapacitation period goes on to exceed seven days, but you still have to keep a record of the over three-day injury. The following details must be recorded.

- Full name, address and occupation of the injured person.
- Date and time of the accident.
- Location of the accident.
- Nature of the injury and how it was caused.

Details of an accident should be recorded by the injured person, but can be completed by any employee. However, some employers insist that records are completed by first aiders, which requires a first aider to be available at all times. This is a company operating procedure and is not strictly necessary. Completion of an accident record does not meet the employer's obligation to report specific accidents and dangerous occurrences to the HSE or Local Authority.

All accident records must comply with GDPR to ensure the confidentiality of entries. The HSE's accident book BL510 enables each accident record to be detached and stored separately when complete, thus maintaining the confidentiality of the injured person's details. All personal information must be kept in confidence, and in a secure location, such as a lockable cabinet. Accident books designed and produced in-house may still be used in the following circumstances.

- Where all details required by BL510 are recorded.
- If the requirements of GDPR are met.
- When information recorded electronically can be made readily available in hard copy.

13.5 Calculating the incidence and frequency rates of accidents

From company accident records and other statistics, it is possible to calculate the incidence and frequency rates for accidents at a particular place of work and for the types of injury, severity or duration.

13.5.1 Accident incidence rate (AIR)

The incidence rate is based on the number of accidents, taken over a fixed period (usually 12 months), per 100,000 employees. The formula highlighted below is the one most commonly used by UK Government departments to calculate the incidence rate.

$$\text{Incidence rate} = \frac{\text{Number of reported injuries in a 12-month period} \times 100,000}{\text{Average number of employees in a year}}$$

If, during a 12-month period, there were six reportable accidents and during that year the company employed an average of 120 employees, the calculation would be:

$$\frac{6 \times 100,000}{120} = 5,000$$

13.5.2 Accident frequency rate (AFR)

The accident frequency rate allows a calculation to be made that balances the number of reportable accidents that occur against the number of hours worked in a fixed period (usually 12 months).

$$\text{Frequency rate} = \frac{\text{Number of accidents in a 12-month period} \times 100,000}{\text{Number of hours worked in that period}}$$

If a company had six reportable injuries in a period during which its 120 workers worked a total of 280,000 hours, the accident frequency rate would be:

$$\frac{6 \times 100,000}{280,000} = 2.14$$

It should be noted that the HSE uses a multiplier of 1,000,000 rather than 100,000 in its guidance on the calculation of AFR. Generally the industry uses 100,000: 100,000 working hours at 45 hours per week for 49 weeks per year represents 45 years' work for one person or 90 years' work for two persons, and so on.

13.5.3 All accident rates

Similarly, by using the above incident and frequency calculations, all accident rates (minors, lost time and reportable accidents) can be calculated. Computers are ideal for calculating the various incidence and frequency rates, moving averages, and so on.

There is a wide range of commercially-available computer software to manage accident statistics or a simple spreadsheet can be easily devised, using the known labour resources and their total working hours as a basis for calculating the statistics. Accident statistics are often displayed as bar charts, histograms and graphs to visualise trends more clearly.

The robust and effective reporting of accidents, along with their thorough investigation, can have major benefits for a company, such as:

- reduced costs by proactively implementing change and preventing accidents
- identifying training needs, which will also improve performance
- satisfying clients and principal contractors that the workforce is properly trained and safety orientated
- a possible reduction of insurance premiums following years of hard work to reduce accidents.

13.6 Post-accident investigation

An effective investigation will meet the following criteria.

- Be factual and without bias.
- Clearly show the sequence of events leading to the accident or incident.
- Establish the immediate cause(s) and the underlying cause(s) (for example, unsafe acts or conditions).
- Identify the root cause(s) (for example, performance-influencing factors such as lack of safety systems, poor work planning leading to workload pressures, or a poor health and safety culture).

The root cause(s) must be identified, in order to be able to learn lessons from accidents and incidents, and then take action to prevent re-occurrences.

13.6.1 Accident procedure

Many companies have established procedures to be followed in the event of an accident.

The procedure below is given as general guidance and outlines the steps that should be taken immediately after an accident.

1.	Identify any injured persons and assess the severity of the injuries.
2.	Make the area safe, preserve the scene and notify relevant parties (such as management, the health and safety team or HSE).
3.	Call the appropriate emergency services, and arrange first aid for the injured person(s) if safe to do so.
4.	Isolate machinery, tools or equipment if safe to do so.
5.	Do not disturb or move anything unless to release an injured person.
6.	Ensure that any remaining hazards are controlled.
7.	Take notice of anything significant and make general observations at the scene of the accident.

13.6.2 Conducting an investigation

It is not usually practical to investigate every minor accident, but those involving specified or serious injuries to persons and major damage to plant or equipment should be thoroughly investigated so that immediate action can be taken to prevent a recurrence.

The following steps may be useful as a guide to the steps to be taken.

- Investigate promptly and record evidence.
- Identify types of evidence (for example, factual or corroborative).
- Interview the injured person, if possible.
- Question the person in charge and other supervisors.
- Obtain details of the injured person's job and what they usually do.
- Interview witnesses.
- Inspect plant, equipment and tools for signs of misuse or defects.
- Establish the full sequence of events.
- Ascertain the nature and extent of the injury or damage.
- Complete the accident report and the accident book.
- Notify the appropriate authorities.

ACCIDENT REPORTING AND INVESTIGATION

13.6.3 Investigating promptly

The sooner an investigation is started, the better – provided it is safe to do so. Engineers and supervisors will be anxious to find ways to repair any damage to plant, machinery or buildings, but the **first priority** should be to establish the cause(s) of the accident. Safety specialists, managers and safety representatives should concern themselves solely with the safety implications and preventing a recurrence. It is important that the investigation is properly supervised and organised. Where the police or HSE inspectors need to investigate, any other persons responsible for, or involved in, investigating the accident must take care not to disturb possible evidence at the scene.

13.6.4 Recording evidence

Statements from witnesses should include their age and occupation. The time, date and place of interview should be indicated at the end of the statement. Witnesses' statements should always be written in their own words, even if these include slang or expletives.

The completed statement should be read to the witness and signed by them and by the person who took the statement.

13.6.5 Identifying the types of evidence

The following will usually be included within evidence.

- Statement of witnesses and others given orally, or in writing. 'Others' may include experts who, for example, might have been called in to examine a machine or the state of a scaffold.
- Documentation of all kinds.
- Material exhibits of all kinds.

Factual evidence comprises the facts related by persons directly involved, and by witnesses who are able to say what they felt, saw, heard, or give an expert opinion. This type of evidence is primary, direct and positive and should be written in simple language, keeping to the facts and avoiding inferences, opinions and beliefs. The facts should be recorded clearly, accurately and in sequence.

The best witnesses are those persons directly involved, who have the following qualities.

- Can listen carefully to the questions.
- Are able to answer directly, fairly, impartially and truthfully.
- Can state clearly when they do not know the answer.
- Are able to remain calm when they are being asked questions.

Material evidence includes, for example, items of plant, equipment, machines, scaffolds, ladders or hand tools, where the use of, or the state or condition of, the item has a bearing on the accident.

Corroborative evidence tends to support the truthfulness and accuracy of the evidence that has already been given. The confirming evidence may take the form of site records, plant or maintenance records, warning notices and written procedures or reports made by safety officers or safety representatives.

Opinions are not generally acceptable as evidence in a court of law, but people in the vicinity of an accident should be asked to give an opinion. In this way a full picture can be built up of the circumstances of the accident.

Experts, or specialists, who are familiar with the type of accident, or technical and other factors surrounding the accident, may be called upon to express their **expert** opinions. When there is a lack of real or factual evidence, other forms of evidence (such as circumstantial and corroborative evidence) tend to become more valuable.

Photographs taken immediately after an accident record the state of the scene and often highlight conditions that existed at the time. The position of machines, equipment and tools, obstructions, and factors such as floor conditions, space and dimensions, may show up well on photographs.

The time, date and place or subject photographed should be recorded on the photographs.

Too many photographs are far better than too few, and it is a good idea to make drawings of the area where the incident happened. Digital photography may not be accepted as primary evidence but may be suitable as supportive evidence.

Procedures should be in place to ensure that photographs have not been, or cannot be, altered in any way, as this would destroy their value as evidence.

13.6.6 Interviewing the injured person

In the event of a major incident or fatality, the police may initially take charge to investigate or rule out any criminal intent. They should be notified immediately of any intention to interview injured persons, as the nature and content of interviews may form part of their criminal investigation. Once the police have satisfactorily ruled out any criminal intent, the HSE will then take over the investigation.

Interviewing the injured person should be an early priority. Even the briefest description of the accident should suffice initially.

The physical and mental state of the injured person will need to be considered, and tact and patience will be required during the interview. The injured person should be fit to answer questions.

The injured person should be encouraged to talk about how the accident happened and it is important they have confidence and trust in the listener. It is important to stress that the purpose of the investigation is to find the cause so that preventive action can be taken. Blame should not be apportioned.

Questioning should not take the form of an interrogation. Someone well known to the injured person is probably the best person to do this. Safety officials are more likely to receive the co-operation of an injured person if they are able to demonstrate a genuine interest in their welfare and recovery. This may involve visiting the injured person, with the doctor's approval, in hospital or at home.

Given the nature of the industry, often with multiple layers of sub-contracting, there may be a number of interested parties. Sharing statements is one way of reducing the stress on witnesses. However, regulators (such as the police and HSE) may be less forthcoming about such arrangements as they may see this as an employer trying to interfere in a criminal investigation.

13.6.7 Questioning the person in charge

Establish from the injured person, manager, supervisor or the person in charge what the normal job and tasks of the injured person were. Did they include the activity that led up to the accident?

? Other questions that might be asked

- What task or type of job was being performed?
- Was it planned or part of a planned activity?
- At what stage of the work did the accident occur?
- Was the person involved trained and authorised and, if so, when?
- Was the person authorised to be where the accident occurred?
- What instructions had been given?
- Were safe and correct procedures being followed?
- Did unsafe acts cause the accident? If so, were they those of the injured person, workmates or others?
- Did any unsafe conditions contribute to the accident?
- What safety equipment or personal protection was available and in use?

13.6.8 Interviewing witnesses

Skill is required when interviewing. Witnesses should be interviewed one at a time. If they wish to say anything before notes are taken, they should be allowed to do so.

? Basic questions interviewers should seek answers to

- What did the witness actually see or hear?
- What was the witness doing at the time?
- What was the proximity of the witness to the accident or occurrence?
- What actions did the witness take?
- What actions did others take before and after the accident?
- What was the condition of the workplace at the time?
- What hazards or unsafe conditions existed and what unsafe acts were performed?
- What was the probable cause(s) of the accident or occurrence?

Skilled interviewers allow witnesses to tell things in their own way, intervening only to clear up specific points or answers where necessary. Questions should be impartial, they should not be leading, and the interviewer must be unbiased. Questions and answers given should be recorded.

It is quite acceptable to go through an incident with a witness making rough notes and then to take a statement after that. That way, the witness often has more chance to remember the facts, and sometimes provides far more detail on the second run through.

The important witnesses are those persons who were involved. Their evidence will be more valuable than evidence from witnesses who saw or heard only from a distance, although they, too, should be interviewed. Corroborative evidence and information is often required, particularly when witnesses are few or are not reliable. As much evidence and information as possible should be collected, since the action taken to prevent a recurrence will be based on what is learned.

ACCIDENT REPORTING AND INVESTIGATION

13.6.9 Inspecting plant for misuse or defects

Inspection of plant, equipment, tools and machinery immediately after an accident may reveal signs of misuse, or defects, which may or may not have contributed to the accident.

The scene should also be carefully examined to see if trip hazards, slippery floors, or some other defect contributed to, or caused, the accident.

Assistance from specialists and persons directly involved or familiar with the type of plant, equipment, tools or machinery in question can provide valuable information.

13.6.10 Establishing a sequence of events

Evidence gained from interviews and from inspection of the scene, plant, equipment, tools or machinery should give an indication of the sequence of events leading up to the accident.

13.6.11 Ascertaining the extent of injury or damage

It is not always possible to ascertain the full extent of injuries and damage resulting from an accident.

There may be complications or delayed effects from injuries. The total time off work will obviously not be known at the time of investigation.

Whilst it may be easy to identify the extent of the damage caused to plant, machinery, equipment, buildings and materials, it is far from easy to measure the overall effects of the accident in terms of lost time, lost production and, of course, the suffering of the injured person or persons.

13.6.12 Completing the accident investigation report

Accident report details will vary, depending on who produces the report and whom the report is for. To help eliminate or reduce this variation, guidance in making reports and the use of a standard form is recommended.

As far as possible, reports should be concise, based upon fact rather than speculation, be unbiased and should summarise the essential information obtained during the investigation.

It is becoming relatively common for HSE inspectors to simply ask for a copy of an accident investigation report completed by the employer. It is a criminal offence to intentionally falsify or manipulate the findings of an accident investigation report and it is therefore important to make sure that the report is unbiased and, as far as possible, accurate.

13.6.13 Lessons learnt

It is essential that, following an investigation, lessons learnt are communicated throughout relevant areas of the business, and with wider industry groups if appropriate.

Changes to controls, systems, procedures, communications and so on should be implemented to ensure the continual improvement of health and safety.

 **For further information on investigating accidents and incidents visit the HSE website.**

Appendix A – Accident/incident reporting matrix

Type of incident	Enter in company records and investigate	Put in accident book	Send Form 2508 to enforcing authority*	Phone enforcing authority and send Form 2508*	Send Form 2508A to the enforcing authority*
Any incident or near miss	✓				
Every injury**	✓	✓			
Over three days' time lost	✓	✓			
Over seven days' time lost ***	✓	✓	✓		
Specified injury or fatality	✓	✓	✓	✓	
Dangerous occurrence	✓		✓		
Reportable occupational disease*	✓				✓

* Refer to the accident reporting section of this chapter for the different methods of submitting accident reports to the enforcing authority.

** Note employers and others with responsibilities under RIDDOR must still keep a record of all over three-day injuries. An accident book record will be enough.

*** Note the deadline by which the over seven-day injury must be reported is 15 days from the day of the accident.

Index



**Scan the QR code to complete a brief survey,
and share your feedback on the course.**

INDEX

- accident book 97, 135, 138, 159-161
- accident records 151, 160, 162
- accident reporting
 - calculating incidence and frequency rates 160
 - how to report 159
 - legislative requirements under RIDDOR 156-9
 - near misses 147
 - out of hours 159
 - overview 156
 - post-accident investigation 161-4
 - remedial actions 130
- accident statistics 50, 52, 84 131, 146, 148, 160
 - see also Bird's triangle
- accidents
 - causes of 149
 - costs of 148
 - and hazards 151-2
 - and inexperience 152-3
 - and non-workers 158
 - investigation of 161-4
 - prevention and control 146-154
 - procedures 161
 - supervision and control 153
 - trends 148
 - witnesses to 163
 - see also accident reporting
- ALTE work statements 101-102
- Approved Codes of Practice (ACoPs) 4, 5, 8, 19, 127, 130, 139, 141-2
- asbestos 3, 6, 21, 38, 40, 68, 70, 90, 94, 97, 135, 139, 140, 152, 158
 - see also *Control of Asbestos Regulations 2012*
- assembly points 90, 91, 104, 135
- audits 126-132

- behavioural competence 87
- behavioural safety 105-118
 - programme 111-4
 - see also human factors
- Bird's triangle 146
- burden of proof 9
- buried services see underground services

- card schemes 86-7, 91
- CDM see *Construction (Design and Management) Regulations 2015*
- CITB
 - Health, Safety and Environment test* 88
- clients
 - definition 35, 44, 45
 - duties of 35, 45
 - and health and safety file 35
 - see also domestic clients
- clothing 92, 94, 136, 142
- Codes of Practice see Approved Codes of Practice (ACoPs)
- cofferdams and caissons 42, 136, 137, 140
- communication
 - and behavioural safety 115-6
 - and health and safety management 59
 - with non-English speaking workers 85, 77, 100-104
- competence
 - and behavioural safety 87
 - definition 51
 - and training 90
- competency card schemes 86-7, 91
 - see also card schemes
- competent persons 51-2
- compliance, standards of 8-9
- confined spaces 90

- Construction (Design and Management) Regulations 2015 (CDM)*
 - 30-48
 - consultation with workers 33
 - emergency procedures 42
 - introduction and aims of 30
- construction phase 33, 40-1, 44
 - see also notifiable projects; pre-construction phase
- construction phase plan 33, 40-1
 - work involving risks 43
- consultation with workers
 - under CDM 33
 - and health and safety management 57, 59
 - and induction 90
 - and risk profiling 57
- contractors
 - and accidents 153
 - appointment and management of 60-1
 - assessment of 64-5
 - definition 38
 - duties under CDM 33, 37-9, 123
 - see also principal contractors
- Contractors Health and Safety Assessment Scheme (CHAS) 64-5
- control measures 57
- Control of Asbestos Regulations 2012* 3, 6, 139
- Control of Noise at Work Regulations 2005* 21, 142
- Control of Substances Hazardous to Health Regulations 2002*
 - 8, 30, 91, 134, 135, 139, 140, 149
- Corporate Manslaughter and Corporate Homicide Act 2007* 3, 11, 14
- criminal liabilities 2
- critical safety behaviours 111-3

- danger areas 135, 139
- dangerous occurrences 7, 91, 136, 138 156-9
- dangerous substances 135, 139
 - see also hazardous substances
- Dangerous Substances and Explosive Atmospheres Regulations 2002*
 - 139
- deaths see fatalities and injuries
- demolition 41, 87
- designers
 - definition 21, 44, 45, 36
 - duties of 21, 34, 36-7, 45
 - see also principal designers
- diseases 7, 156-9
- documentation see record keeping, health and safety
 - management systems 52-3
- domestic clients 35, 44, 45
 - see also clients
- duty holders
 - under CDM 6, 31, 32, 33, 34, 35, 45-6
 - under HSWA 20-2
 - under the *Work at Height Regulations 2005* 7

- electrical equipment
 - documentation 135
 - inspections 137, 139
 - and training 91
- Electricity at Work Regulations 1989* 6, 139, 140
- emergency
 - procedures 42, 81, 91
 - routes and exits 42
- employees' duties 6, 21
- employers' responsibilities
 - under HSWA 5-6, 20
 - induction training 85, 97
 - risk assessments 69

- Employment Medical Advisory Service (EMAS) 19, 25
- enforcement of health and safety legislation 22, 23, 25-6
- Equality Act 2010* 124
- equipment checks
 - user checks 135, 137
 - visual checks 135
- escape routes *see* emergency routes and exits
- evacuation *see* emergency procedures
- excavations 42, 92, 136, 137, 140
- explosive atmospheres 135, 139
- explosives 41, 136, 140

- falsework 136, 140
- fatalities and injuries
 - reporting and investigation 7, 138, 156-164
 - statistics 2-3, 146
- fire 92, 136, 140
 - detection 42
 - fighting appliances *or* equipment 42, 137
- first aid 92, 136, 140
- flammable liquids 92
- fragile surfaces 136, 140

- gas *or* gases 135, 138, 139, 158-9
 - fittings 138
 - incidents 159
- General Data Protection Regulations* 135, 138, 156, 160

- hazardous substances 81, 91, 136, 140
- hazards
 - identification of 73, 151-2
- Health, Safety and Environment test* 87-8
- Health and Safety at Work etc. Act 1974 (HSWA)* 2-3, 4, 5, 8, 9, 12, 13, 18-27, 51, 54, 60
 - prosecutions under 22-4, 26-7
 - requirements of 20-2
 - and safety representatives 9, 19, 20, 23, 122
- health and safety committee 92
- Health and Safety (Display Screen Equipment) Regulations 1992* 6-7, 142
- Health and Safety (Enforcing Authority) Regulations 1998* 25-6
- Health and Safety Executive (HSE)
 - accident costs 149
 - accident statistics 146
 - behavioural safety 106
 - enforcement 22, 23-4
 - fees for intervention 25
 - health and safety management systems 50, 53, 66
 - inspections 129
 - powers and duties 22-4
 - prosecutions 22, 26-7
 - risk assessments 73-4
 - and worker engagement 78, 121
- Health and Safety (Fees) Regulations 2012* 3, 25
- Health and Safety and Nuclear (Fees) (Amendment) Regulations 2023* 3, 25
- health and safety inspections *see* audits; inspections; inspectors
- health and safety law 2-15
- health and safety management systems 49-66
 - competent persons 51-2, 55, 60
 - delivery 56-62
 - documentation 61
 - economic case for 52
 - example of health and safety policy 66
 - HSG65 (*Managing for health and safety*) 50, 51, 52, 53
 - ISO 45001:2018 50, 52-3
 - monitoring and review 63
 - overview 50-2
 - planning 53-6
 - reviewing 63
 - risk profiling 56-7
 - third-party contractor assessment 64-5
- Health and Safety (Offences) Act 2008* 3, 12, 24
- health and safety policies 50-1, 53-5, 66, 136, 141
- health and safety and regulations 6-8
- Health and Safety (Safety Signs and Signals) Regulations 1996* 139, 142
- health and safety training *see* training
- health surveillance 92, 136, 141
- hearing protection zones 92, 136, 142
 - see also* noise
- Highways Act 1980* 139
- holes in floors and surfaces 136, 141
- HSG65 (*Managing for health and safety*) 50, 51, 52, 53
- human factors 106, 107, 108
 - see also* behavioural safety

- ill-health *see* diseases
- improvement notices 23-4
- induction
 - content 86
 - difficulties 85-6
 - requirements 84-5
 - training 84
- injuries *see* fatalities and injuries
- inspections 127
 - and reports 129
 - statutory or recommended 137
 - see also* monitoring; user checks; visual checks
- inspections and audits 125-132
- inspectors, powers of 22-3
- instructions *see* training
- insurance (employers' liability) 136, 141
- ionising radiation 135, 136, 138, 141
- ISO 45001:2018 50, 52-3

- job design 108

- language difficulties 78, 85
- latent failures 114
- lead and lead work 136, 141
- Legal Aid, Sentencing and Punishment of Offenders Act 2012* 4, 12, 24
- legislation *see* health and safety law
- lifting
 - equipment 8, 93
 - operations 8, 136, 141
- Management of Health and Safety at Work Regulations 1999* 7, 21, 26, 27, 31, 70-2, 84, 122
- manual handling 93, 136, 141
- manufacturers' duties 20-2
- medical advice *see* Employment Medical Advisory Service (EMAS)
- mentoring 116
- method statements 69, 78, 80-81, 93
- monitoring
 - active and reactive 62, 127
 - of health and safety management systems 62
 - see also* user checks; visual checks

- National Occupational Standards (NOS), for management and leadership 122
- near-miss incidents 69, 147

INDEX

- noise 94, 136, 142
 - see also hearing protection zones
- notifiable projects 44
- notification
 - of construction projects 34, 135, 138
- occupational diseases see diseases
- occupational health 18, 81, 94
- occupiers 81
- Occupiers' Liability Act 1957* 10-1, 152
- offences and penalties 11-3, 22-4
- organisation, and behavioural safety 108-9
- penalties see fines; offences and penalties
- performance measurement see monitoring
- permits to work 79, 80, 81
- personal protective equipment 81, 94, 142
- plant and equipment
 - inspection and records 80, 135-7
 - reporting defects 94
- post-construction phase 33
- 'practicable' (definition) 8
 - see also 'reasonably practicable' (definition)
- pre-construction phase 33, 35, 36, 45
- pressure vessels 136, 142
- pre-start assessments 101-2
- prevention principles 31-2, 72
- principal contractors 37-8, 44, 45, 51
 - see also contractors
- principal designers 33, 36, 44, 45
 - see also designers
- prohibition notices 12, 19, 22, 23-4
- project documents, under CDM 39-41
- Provision and Use of Work Equipment Regulations 1998*
 - 7, 134, 137, 142-3
- 'reasonably practicable' (definition) 9
 - see also 'practicable' (definition)
- record keeping 134
 - and inductions 88
 - and risk assessments 74
 - see also accident books; health and safety files; statutory forms
- registration schemes 94
- Regulatory Reform (Fire Safety) Order 2005* 20, 22, 137, 140
- remedial action 130
- reporting
 - of accidents and incidents 156-164
 - of dangerous occurrences 156-9
 - of defects 94
 - of disease 135, 138, 156-9
 - of gas incidents 156
 - how to report 156-9
 - of injuries 156-9
 - of risk control measures 57, 61-2, 74
- Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013 (RIDDOR)* 7, 130, 136, 138, 156-9, 165
- reviewing
 - of critical behaviours 114
 - of health and safety management systems 53
- RIDDOR see *Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013 (RIDDOR)*
- risk assessments 68-81
 - communication 77-8
 - documentation 79-81
 - legislative requirements 69-72
 - in practice 76-7
 - principles of 72
 - qualitative assessments 74
 - quantitative assessments 74, 75-6
 - records 74, 141
 - and training 95
 - types of 74-6
- risk or risks
 - and construction phase plan 43
 - control plans 61
 - definition 68
 - operational 152
 - profiling 57
- safe places of work 5, 41, 58, 130
- safe systems of work 95
- Safecontractor (health and safety assessment scheme) 65
- Safemark (health and safety assessment scheme) 65
- safety representatives
 - and inspections 127
 - statutory forms 135, 136, 138, 142
- Safety Representatives and Safety Committees Regulations 1977*
 - 9, 59, 122, 123-4, 127, 138, 142
- Safety schemes in procurement (SSIP) 64, 65
- safety surveys 128
- safety tours 127-9
- scaffolding 95, 136, 142
- security see site security
- sentencing guidelines, for health and safety offences 11
- signs and notices 95, 136
- site
 - inductions 84-98, 104
 - induction topics 90-6
 - layout 95, 104
 - plans 104
 - security 95, 152,
- statutory forms 134-144
- statute law 5
- supervisors
 - actions for 77, 80
 - and competence 87
 - and health and safety management 58
- suppliers' duties 21-2
- third-party contractor assessment 64-5
- toxic materials/substances see dangerous substances; hazardous substances
- traffic routes 42, 96
- training
 - and health and safety management systems 60
 - and time off, and provision of facilities 10
 - induction training 84-98
 - language difficulties 85-6, 100-1
 - plant inspection and operator 80
 - promotion of 89
 - records 88
 - use of images 102-4
 - see also coaching; information provision; mentoring
- underground services 80, 96, 151
- vibration 21, 58, 70, 71, 96, 146, 152, 158, 159
 - whole-body vibration (WBV) 96
- waste disposal 96, 143
- welfare facilities 47-8, 96
- Work at Height Regulations 2005* 3, 7, 30, 137, 140, 141, 142, 143
- worker engagement 78, 120-4

working at height 70, 96, 136, 143
Working Time Regulations 1998 7, 143

SITE SAFETY PLUS

Construction site health and safety

A: Legal and management

This section contains a framework of information needed to understand what the law requires and the main legislation that applies to construction work.

It details how to manage health and safety in a business, the methods of assessing risk and developing safe systems of work, accident prevention and investigation, plus some ways to monitor and maintain the controls needed on site.

This publication is also the official reference material for the Site Safety Plus *Director's role for health and safety* (DRHS®), a one-day course for construction company directors.

DRHS is a registered trade mark of the Construction Industry Training Board.